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Paleomagnetic Study of the Eastern Klamath Terrane, California, and Implications for the Tectonic History of the Klamath Mountains Province

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Paleomagnetic study of Permian through Jurassic volcanic and sedimentary strata of the Eastern Klamath terrane has shown the remanent magnetization of many of these rocks to be prefolding and most likely primary. Similarities in magnetic declinations recorded by coeval strata over a broad area are consistent with the hypothesis that the terrane, in general, has behaved as a single rigid block. Paleomagnetic data indicate that the volcanic island arc represented by this terrane, the nucleus of the province, was facing toward the present southwest during late Paleozoic time, although its orientation during earlier periods is unknown. Whether the arc was separated from the North American craton by a small marginal basin or originated far offshore cannot be determined from paleomagnetic data. The declination anomalies for both Permian and Triassic strata are similar (average = 106° ± 12°), so we infer that clockwise rotation of the late Paleozoic arc did not begin until latest Triassic or earliest Jurassic time. The arc may have completed its initial rotation with respect to stable North America by Middle Jurassic time. After some retrograde motion, the arc was again facing west by the Late Jurassic, by which time some of the more westerly terranes of the province had become attached to the Eastern Klamath terrane. The composite Klamath Mountains terranes continued to rotate until the final 60° of clockwise rotation was nearly complete by the Early Cretaceous. Coincidence of the waning stages of rotation, at about 136 Ma, with the beginning of deposition of the basal Great Valley sequence onto the Klamath basement probably represents the completion of accretion of the Klamath Mountains terranes to the North American continent. Nearly all the rotation occurred while the Klamath Mountains terranes were part of a converging oceanic plate, with only about 20° of rotation in mid-Tertiary time during Basin and Range extension. No data currently available show evidence for any significant latitudinal displacement of any Klamath Mountains terranes relative to cratonic North America.

INTRODUCTION

The Pacific Coast of North America consists of numerous fragments (terranes) of oceanic crust and island arcs that accreted to the western margin of the craton mainly during the Mesozoic but also during the Cenozoic. Many of these fragments are strikingly dissimilar to their adjacent neighbors, suggesting that they did not form in close proximity to each other. Many have yielded paleomagnetic and paleontologic evidence that point to great latitudinal displacements, and many have undergone large clockwise rotations. Some have been disrupted by faulting subsequent to accretion, and their fragments are now distributed along the continental margin. The Klamath Mountains province of northern California and southern Oregon, an arcuate structure composed of several such terranes, is one of a number of elements that form the tectonic mosaic of the western Cordillera (Figure 1). The west side of the composite Klamath Mountains terrane is underthrust by rocks of the Coast Ranges, but the suture between the east side of the Klamath Mountains and the basement rocks of adjacent provinces is obscured beneath a cover of Cretaceous and Cenozoic strata. The Klamath Mountains figure prominently in various tectonic reconstructions of the Pacific Coast region (see, for example, *Grommé et al.* [1986] and *Wells and Heller* [1988]), and paleomagnetic data could provide valuable constraints to the various tectonic models. Among questions to be answered is whether any of the Klamath Mountains rocks indicate a paleolatitude anomaly or other significant tectonic disruption. The arcuate shapes of the province and many of its formations also have caused many to speculate whether these are primary features or the result of internal deformation, i.e., oroclinal bending. Finally, paleomagnetic data may be able to provide indirect evidence about possible relations of the province to coeval arc volcanics in the adjacent areas. In order to provide firsthand paleomagnetic information from the Klamath Mountains that could aid in the understanding of the accretionary process and the tectonic history of this area, we initiated a paleomagnetic investigation of the easternmost of the terranes of the province, the Eastern Klamath terrane.

The Eastern Klamath terrane consists of a sequence of Paleozoic and Mesozoic island arc deposits that were built upon oceanic crust represented by the Trinity ultramafic sheet [*Irwin*, 1977]. Isotopic age determinations [*Lanphere et al.*, 1968; *Mattinson and Hopson*, 1972; *Jacobsen et al.*, 1984; *Wallin et al.*, 1988] indicate that much of the Trinity ultramafic sheet is Ordovician in age, although some Early Cambrian dikes are locally present [*Wallin et al.*, 1988]. A large central exposure of the Trinity ultramafic sheet separates the overlying strata into two dissimilar areas. Structurally complex strata of the Yreka-Callahan area are Ordovician to Devonian in age [*Potter et al.*, 1977]. In contrast, Lower Devonian through Middle Jurassic strata [*Sanborn*, 1960; *Kinkel et al.*, 1956; *Albers and Robertson*, 1961] of the Redding section, where our paleomagnetic sample localities are located, are a deformed but coherent, generally southeastward dipping sequence. Even here, however, the rocks often are poorly exposed, and detailed mapping is incom-

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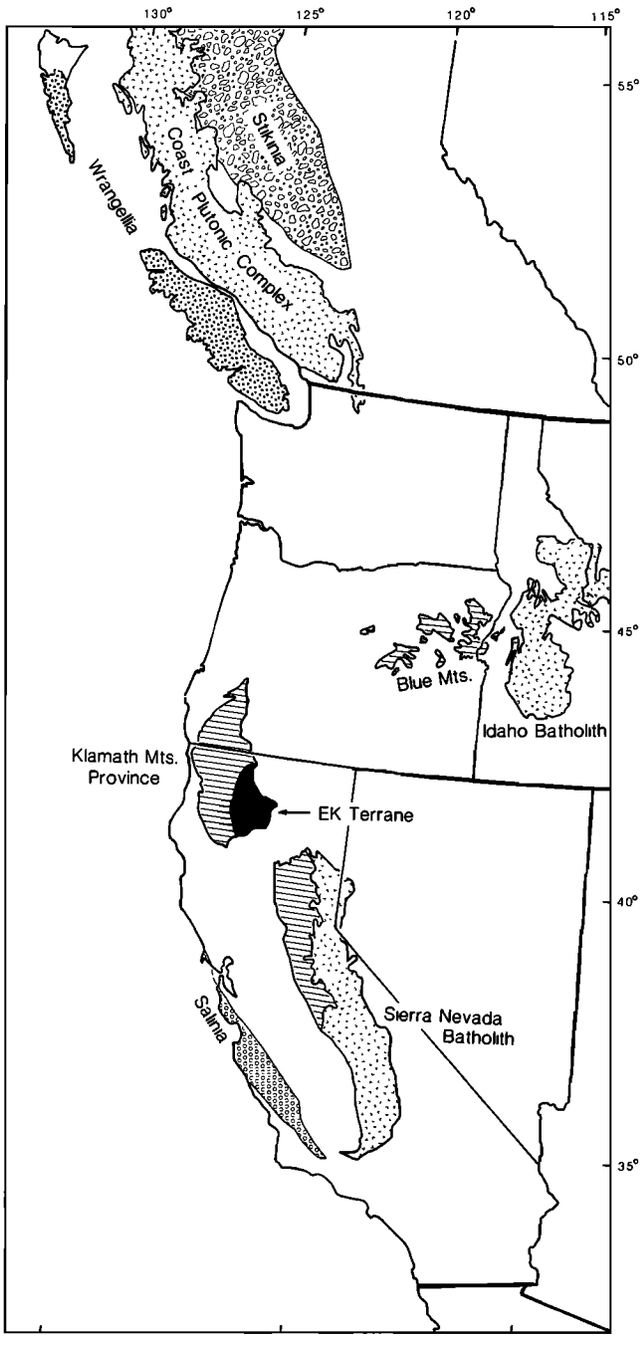


Fig. 1. Sketch showing location of the Klamath Mountains province with respect to selected elements of the western Cordillera. Modified from *Monger and Berg* [1987], and *Silberling et al.* [1987].

plete. The rocks have been subjected to conditions of metamorphism ranging from zeolite to low greenschist facies [*Kinkel et al.*, 1956; *Curtis*, 1983; *Coleman et al.*, 1988; *Renne et al.*, 1988; *Renne and Scott*, 1988; *Miller*, 1989]. Penetrative deformation was not seen at any of the outcrops that we sampled.

Preliminary results of our investigation were described by *Irwin et al.* [1984] and *Mankinen et al.* [1984, 1988]. Other paleomagnetic investigations of rocks in the Redding section were conducted by *Achache et al.* [1982], *Fagin and Gose* [1983], and *Renne et al.* [1986, 1988].

UNITS SAMPLED

The paleomagnetic samples were obtained principally from the Lower Permian McCloud Limestone and younger formations along two widely spaced transects shown in Figure 2. One of the transects extends about 5 km eastward from the McCloud Arm of Shasta Lake and is mainly along Nosoni Creek. The other transect extends nearly 16 km southeastward from Lake McCloud and is mainly along Hawkins Creek. Figure 3 contains diagrammatic stratigraphic columns showing the formations and approximate stratigraphic positions of paleomagnetic sites at both transects. Also sampled were the Permian Pit River stock (PR1) and the Devonian Mule Mountain stock (MM1) at the localities shown in Figure 2. Paleomagnetic results from some Early Cretaceous postamalgamation plutons that intrude the Eastern Klamath terrane are described in a separate report [*Mankinen et al.*, 1988].

The stratigraphic section of Permian rocks exposed along Nosoni Creek was measured and described by *Coogan* [1960]. There the McCloud Limestone appears to be conformably overlain by thin interbedded gray limestone, siltstone, tuff, and black chert assigned to the Nosoni Formation. A similar sequence of rocks is seen at the contact along the west side of Lake McCloud. Although the contact seems conformable at both transects, faunal evidence [*Skinner and Wilde*, 1965] implies that there could be a significant hiatus between the Lower Permian McCloud Limestone and the Upper Permian Nosoni Formation in some areas. The Nosoni Formation is approximately 1900 m thick at the type locality along Nosoni Creek [*Coogan*, 1960] but thins greatly toward the north and is only 316 m thick in the Lake McCloud region [*Miller*, 1989].

The Dekkas Andesite is approximately 1200 m thick at Nosoni Creek [*Coogan*, 1960], and pyroclastic rocks of this unit constitute the bulk of the Permian section in the vicinity of the Hawkins Creek transect where the formation reaches a thickness of about 5500 m [*Miller*, 1989]. While the Nosoni Formation has yielded numerous fossils, few fossils have been found in the lower part of the Dekkas Andesite; they establish the age of the Dekkas as early Guadalupian [*Coogan*, 1960] (Late Permian [*Furnish*, 1973]). The possibility that the Dekkas Andesite may be partly Triassic in age has, therefore, been considered by various researchers. At the Hawkins Creek transect, a bed of corals that was engulfed by pyroclastic deposits is exposed approximately 1,500 m above the base of the Dekkas Andesite. These corals, although poorly preserved, appear to belong to the long-ranging genus *Stereostylas* (C. H. Stevens, written communication, 1983). A thick limestone interbed near the top of the Dekkas Andesite along Hawkins Creek contains abundant large fusulinids that R. C. Douglass (written communication, 1980) considers to be early Guadalupian (Late Permian) in age. The same limestone interbed contains fragmented bryozoan colonies with partly obliterated skeletal microstructures. Although a specific age could not be determined on the basis of the bryozoans, O. L. Karklins (written communication, 1981) notes that the assemblage is typical of the late Paleozoic. Three other limestone lenses at about the same stratigraphic position in the Dekkas as the fusulinid-bearing interbed and between 1.5 and 9 km southeast of our locality yielded a new species of a Permian waagenophyllid coral, but of a genus previously considered

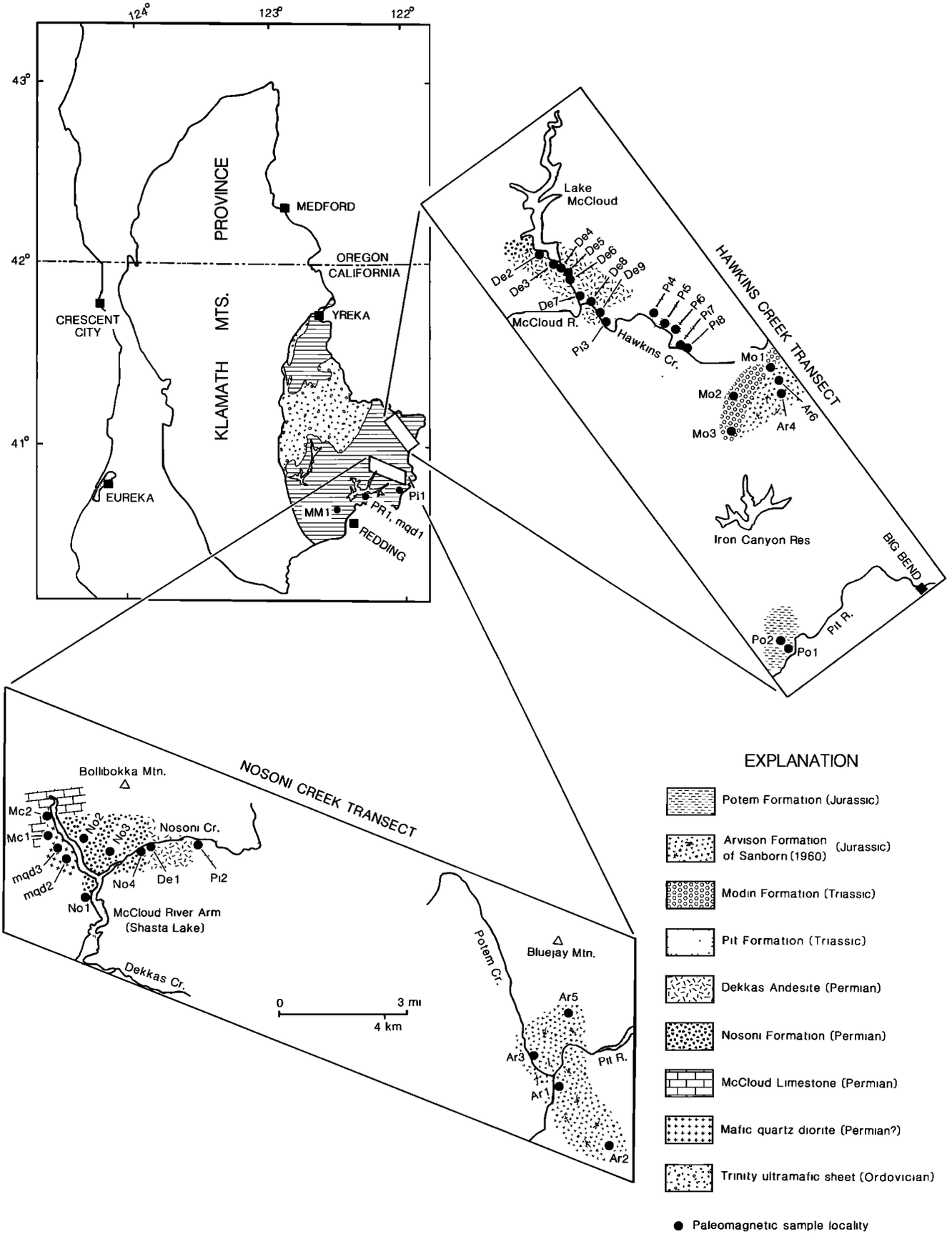


Fig. 2. Sketch map of the Klamath Mountains province showing paleomagnetic sampling sites in the Eastern Klamath terrane (ruled area). Map patterns indicate formations sampled at each locality and are not intended to accurately portray the geology. See Table 1 for site identifiers. Modified from *Mankinen et al.* [1984].

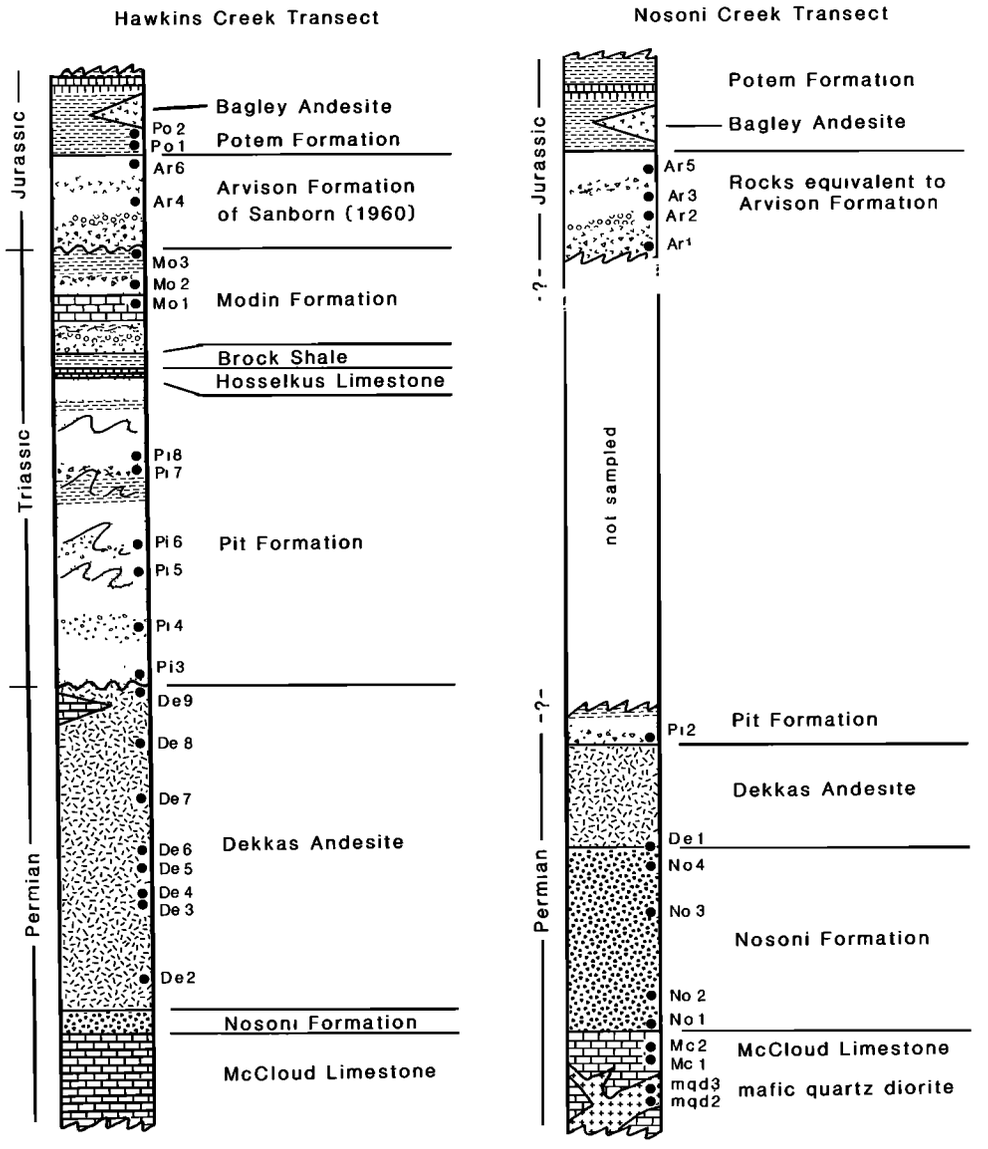


Fig. 3. Diagrammatic sections of the two transects in the Eastern Klamath terrane showing formations and approximate stratigraphic position of paleomagnetic sample localities. Modified from *Coogan* [1960] and *Sanborn* [1960]. The Arvison Formation of *Sanborn* [1960] has not been mapped at the Nosoni Creek transect, but samples were obtained from rocks believed to be its equivalent (see text).

to be diagnostic of the Tethyan province [*Stevens et al.*, 1987; *Miller and Wright*, 1987]. Thus the paleontologic data generally indicate that the Dekkas Andesite is Permian in age.

Siliceous shales and siltstones of the Pit Formation overlie the Dekkas Andesite with slight angular discordance at the Hawkins Creek transect. The thickness of the Pit along this transect was estimated to be 1065 m [*Curtis*, 1983] although a few broad, open folds make the width of its exposure comparable to that of the Dekkas Andesite. The Pit was estimated to be 600–1500 m thick [*Albers and Robertson*, 1961] near the Nosoni Creek transect. The Pit Formation is generally considered to be Middle and Late Triassic in age. Radiolarians from a shale bed in the lower part of the formation have been tentatively identified as Early Triassic in age, and those from the overlying Brock Shale as Middle Triassic, both of which indicate that the Pit at the Hawkins Creek transect may range from Early to Middle Triassic in age [*Curtis*, 1983]. However, a tuffaceous chert layer at sample locality Pi4, approximately 600 m above the contact with the Dekkas Andesite at Hawkins Creek, has yielded late Middle or early Late Triassic radiolarians (C. D. Blome, written communication, 1981). Also, the Brock Shale is generally considered to be Late Triassic in age. Additional study is required to firmly establish the precise limits to the age of the Pit at Hawkins Creek.

The Pit Formation at the Nosoni Creek transect seems to conformably overlie the Dekkas Andesite [*Coogan*, 1960]. Radiolarians from tuffaceous chert at our sample locality in the basal part of the Pit proved to be Permian in age [*Silberling and Jones*, 1982], which indicates that the Pit was deposited over a longer time period than previously thought. If part of the Pit Formation is Permian, then the Bully Hill Rhyolite must also be Permian rather than Triassic in age and might be a phase of Dekkas Andesite volcanism, as suggested by *Silberling and Jones* [1982]. The Bully Hill

Rhyolite was also considered Permian in age by *Albers and Bain* [1985] based on a probable comagmatic relation with the Permian Pit River stock. These studies affirm that the Dekkas Andesite is entirely Permian in age.

The volcanism that was so prevalent in the Redding section during Late Permian time continued intermittently into the Triassic and Jurassic periods. The region was submerged throughout much of its history as evidenced by the interbedding of volcanic rocks with sediments containing marine fossils. Rocks overlying the Pit Formation in the southern part of the Redding section were originally described by *Diller* [1906], who subdivided these strata into five formations. Although he considered the Dekkas Andesite to be Triassic in age, he did recognize that the youngest unit (Potem Formation) was Jurassic. Rocks younger than the Pit Formation along the Hawkins Creek transect were subdivided into six formations by *Sanborn* [1960], who presented abundant fossil evidence that established an age range of Late Triassic to Middle Jurassic for those units. Our paleomagnetic sampling of the post-Permian rocks was confined to the Upper Triassic Modin Formation, the Lower Jurassic Arvison Formation of *Sanborn* [1960], and the Lower and Middle Jurassic Potem Formation. *Sanborn* [1960] estimated the thicknesses of the Modin, Arvison, and Potem (with the intercalated Bagley Andesite) to be approximately 1680, 1550, and 520 m, respectively.

We begin our presentation of results from these Permian through Jurassic strata at the lowest point in the section where a primary remanence might be expected. An unusual diorite dike that intrudes and engulfs the McCloud Limestone over much of the area is present also at the Nosoni Creek transect. Localities within the McCloud Limestone (Mc1, Mc2), the Nosoni Formation (No1), and the diorite itself (mqd2, mqd3) were sampled west of the McCloud River Arm (Figure 2) in order to determine if the diorite could have remagnetized any of the strata. Paleomagnetic results from these five sites will be deferred to a later section.

METHODS USED

The paleomagnetic samples were collected by use of a portable core drill and were oriented with a magnetic compass. A total of 472 oriented cores were obtained from 39 localities shown in Figure 2. Each core was cut into two or more specimens, and the remanent magnetization of one specimen from each core was measured using a cryogenic magnetometer. Representative specimens from each locality were subjected to alternating field (af) and thermal demagnetization experiments to test the stability of remanence and to ascertain the best method of obtaining a representative magnetization direction for each particular outcrop. The af demagnetization experiments were performed using commercial tumbling and nontumbling specimen demagnetizers. Thermal demagnetization experiments were performed in air using a magnetically shielded oven with an internal field less than 10 gammas. Samples for chemical treatment were prepared using the multicoring method described by *Henry* [1979], were leached with 9 *N* hydrochloric acid, and were then rinsed and allowed to dry in ambient fields of less than 100 gammas.

If all pilot samples from a locality had similar responses to the demagnetization experiments, one specimen from each of the remaining cores was magnetically cleaned at a peak field or temperature chosen on the basis of the behavior during stepwise demagnetization. No single demagnetization step or procedure was appropriate for all samples at many localities, and different techniques had to be applied to individual specimens or groups of specimens. In these cases, a stable endpoint had to be determined for each specimen by analysis of directional changes during the demagnetization experiments and of the vector component diagrams of these data. The stable remanence of all specimens was then used to determine a mean direction of magnetization for each locality. These mean directions and associated statistical parameters are given in Table 1. In many instances, the principal component analysis of *Kirschvink* [1980] was used to analyze those sets of demagnetization vector endpoints that appeared to fall on great circle paths on an equal-area projection. These converging remagnetization circles [*Halls*, 1976] can be used to determine either the primary or secondary magnetization direction in a two-component system, depending on which component is the least dispersed. We were able to use this method to determine the directions of remagnetizing fields that had affected our sample localities. In ideal situations, it should be possible to use converging circles in uncorrected (in situ) space to determine an overprint direction and then structurally to correct the data to produce a second set of circles and determine the primary remanence. In this nearly homoclinal section, however, structural attitudes often are too uniform to produce the necessary dispersion in one component relative to the other in order to fully utilize the method. This lack of structural variation also precluded application of the fold test to measure magnetic stability in most circumstances. We used Bingham statistics [*Onstott*, 1980] to establish confidence ellipses for intersections of converging remagnetization circles and conventional Fisher circular statistics [*Fisher*, 1953] in all other cases.

PERMIAN STRATA

Hawkins Creek Transect

Paleomagnetic sampling of Permian strata at the Hawkins Creek transect was confined to the Dekkas Andesite which, in that area, consists almost entirely of pyroclastic rocks. Many of the exposures are quite massive with no apparent bedding, and paleomagnetic sampling was confined to those localities where the attitude of the rocks could be determined unambiguously. Individual sites do not consist of single cooling units; so at each site, geomagnetic secular variation may have been averaged to some unknown degree. Much of the rock along this section is reddened throughout, presumably due, in large part, to deuteric alteration. A total of 160 cores were obtained from the Dekkas Andesite at seven sites along Hawkins Creek which span a stratigraphic thickness of approximately 3000 m. Many sites were affected to some extent by a normal-polarity overprint, but this was readily removed by either af or thermal demagnetization (Figure 4). Thermal demagnetization showed maximum blocking temperatures between 540° and 560° for most samples, although a few had stable magnetizations up to about 590°C. Changes in direction of magnetization became random upon heating to higher temperatures. These observations indicate that the primary carrier of remanence is magnetite or Ti-poor titanomagnetite.

TABLE 1. Site-Average Paleomagnetic Data for Strata of the Eastern Klamath Terrane, California

Map Number	Site				Uncorrected						Corrected						Corrected VGP		
	Latitude °N	Longitude °E	N/N_0	Treatment	I, deg	D, deg	k	α_{95}	Strike, deg	Dip, deg	I, deg	D, deg	k	α_{95}	J_{nrm}	J_{cl}	Latitude °N	Longitude °E	A_{95}
									Potem Formation										
Po2	41.00	238.03	23/23	af+T	64.6	356.5	92	3.2	029.0	16.0	68.7	033.4	92	3.2	49.300	33.500	64.8	290.6	5.0
Po1	41.00	238.03	21/21	af+T	59.6	353.8	41	5.0	032.0	15.0	64.6	021.1	45	4.8	30.900	13.700	73.9	301.1	7.1
									Arvison Formation of Sanborn [1960]										
Ar6	41.09	238.03	10/10	af	33.9	037.6	140	4.1	130.0	33.0	66.9	034.9	140	4.1	1.130	0.353	64.3	297.1	6.3
Ar5	40.86	237.99	13/13	af+T	52.5	022.2	278	2.5	horizontal		NA	NA	NA	NA	1.360	0.824	70.7	344.7	2.9
Ar4	41.09	238.03	10/10	af+T	47.9	023.0	94	5.0	201.5	34.0	38.8	350.6	125	4.3	0.756	0.317	69.3	083.4	4.2
Ar3	40.84	237.97	7/8	af	64.6	356.1	58	8.0	238.0	16.0	49.9	346.3	58	8.0	1.050	0.665	74.9	109.8	9.1
Ar2	40.81	238.01	7/8	af+T	49.0	000.3	23	13.0	181.0	27.0	42.0	332.9	23	13.0	1.880	0.962	62.0	120.1	13.2
Ar1	40.83	237.98	8/8	af	71.0	004.7	51	7.9	326.0	64.0	13.4	040.9	51	7.9	0.742	0.392	40.2	359.7	6.3
									Modin Formation										
Mo3	41.08	238.01	8/8	af+T	34.0	015.5	171	4.3	042.0	88.0	22.9	078.3	171	4.3	1.230	0.630	16.6	326.5	3.6
Mo2	41.09	238.01	8/8	af	67.2	043.4	31	10.1	035.0	32.0	48.7	089.4	31	10.1	0.622	0.149	19.4	305.1	11.3
Mo1	41.09	238.03	13/13	af	56.1	015.8	28	8.0	038.0	69.0	31.0	091.4	40	6.7	0.473	0.136	09.9	314.4	6.0
									Pit Formation										
Pi8	41.10	237.99	0/9	af+T	unstable magnetization										0.121	N.D.	N.D.	N.D.	N.D.
Pi7	41.10	237.99	0/10	af+T	unstable magnetization										0.126	N.D.	N.D.	N.D.	N.D.
Pi6	41.11	237.98	0/10	af+T	unstable magnetization										0.200	N.D.	N.D.	N.D.	N.D.
Pi5	41.11	237.98	0/8	af+T	unstable magnetization										0.341	N.D.	N.D.	N.D.	N.D.
Pi4	41.11	237.97	12/16	af+T	57.6	046.5	19	10.3	057.0	56.5	33.4	107.3	20	10.0	1.460	0.668	−00.4	303.1	9.2
Pi3	41.11	237.95	8/8	af+T	56.6	048.1	68	6.8	057.0	62.0	27.9	109.1	68	6.8	2.920	1.370	−04.0	304.3	6.0
Pi2	40.91	237.82	6/7	af+T	45.8	010.7	29	12.7	0.0	46.0	24.9	041.5	40	10.7	0.056	0.014	44.4	353.2	9.2
Pi1	40.75	237.94	0/8	af+T	unstable magnetization										0.021	N.D.	N.D.	N.D.	N.D.
									Dekkas Andesite										
De9	41.11	237.95	20/22	af+T	61.3	018.2	33	5.7	050.0	75.0	28.1	112.5	33	5.7	3.740	2.170	Jurassic remag.		
De8	41.12	237.95	10/10	af+T	−09.9	187.2	36	8.2	045.0	105.0	−32.6	247.5	36	8.2	5.160	3.000	−28.4	148.6	7.5
De7	41.12	237.94	18/23	af+T	−27.3	241.9	23	7.5	075.0	47.0	−27.4	267.9	23	7.5	2.470	0.953	−11.1	138.2	6.5
De6	41.13	237.93	20/21	af+T	00.6	229.4	34	5.7	075.0	95.0	−25.6	254.9	33	5.8	80.500	54.000	−20.2	147.4	5.0
De5	41.13	237.93	35/42	af+T	11.1	212.4	22	5.3	075.5	77.0	−37.5	230.4	23	5.2	4.500	1.070	−43.2	157.5	5.0
De4	41.13	237.93	6/7	af+T	08.4	223.3	18	16.2	085.0	74.0	−36.3	241.5	18	16.2	4.820	0.352	−34.4	150.4	15.4
De3	41.13	237.93	10/12	T	−01.7	221.4	49	7.0	060.0	90.0	−18.6	241.8	49	7.0	2.350	0.904	−27.4	159.8	5.7
De2	41.13	237.92	8/8	af+T	−22.1	195.9	36	9.4	068.5	70.0	−55.1	258.8	36	9.4	16.800	7.760	−30.2	125.1	11.7
De1	40.91	237.80	15/15	af	61.7	022.5	17	9.6	040.0	42.5	47.1	081.1	22	8.4	0.034	0.017	Jurassic remag.		
									Nosoni Formation										
No4	40.91	237.79	0/6	af+T	unstable magnetization										0.330	N.D.	N.D.	N.D.	N.D.
No3	40.91	237.78	7/8	T	−24.0	088.0	168	4.7	357.0	20.0	−44.0	088.3	168	4.7	3.430	1.111	remagnetized(?)		
No2	40.92	237.77	0/19	af+T	unstable magnetization										10.700	N.D.	N.D.	N.D.	N.D.
No1	40.90	237.77	6/8	af	−57.4	111.7	198	4.8	310.0	60.0	−34.6	181.6	198	4.8	6.730	5.610	Permian remag.		
									McCloud Limestone										
Mc2	40.92	237.75	9/18	af+T	−86.9	183.3	19	12.1	334.5	53.5	−40.1	240.6	26	10.2	0.047	0.008	Permian remag.		
Mc1	40.92	237.75	8/8	af	−87.4	127.2	52	7.7	320.0	65.0	−25.6	227.2	52	7.7	0.031	0.034	Permian remag.		

TABLE 1. (continued)

Map Number	Site Latitude °N	Site Longitude °E	N/N_0	Treatment	Uncorrected I, deg	Uncorrected D, deg	Uncorrected k	Uncorrected α_{95}	Uncorrected Strike, deg	Uncorrected Dip, deg	Corrected I, deg	Corrected D, deg	Corrected k	Corrected α_{95}	J_{nrm}	J_{cl}	Corrected VGP Latitude °N	Corrected VGP Longitude °E	A_{95}
									Mafic Quartz Diorite										
mqd3	40.92	237.75	11/13	af+T	−65.4	123.8	14	12.5	323.0	46.5	−51.5	205.9	37	7.6	6.360	3.210	uncertain structural correction		
mqd2																			
mqd1	40.75	237.69	11/11	af+T	−61.2	193.7	23	3.7	005.0	60.0	−22.0	244.1	23	3.7	5.250	1.090	Planes solution		
									Pit River Stock										
PR1	40.75	237.68	12/12	af+T	59.7	028.0	57	5.8	005.0	60.0	15.1	066.2	57	5.8	9.330	4.017	Jurassic remag.		
									Mule Mountain Stock										
MM1	40.67	237.50	6/8	af	−30.2	193.8	47	9.8	045.0	60.0	−39.7	241.0	47	9.8	0.020	0.005	Permian remag.		

N/N_0, number of samples used in the analysis/total number of samples collected; treatment, alternating field (af) and thermal (T) demagnetization; I, D, inclination, positive downward and declination, east of north, respectively, of remanence vector uncorrected and corrected for attitude of bedding; k, precision parameter $(N - 1)/(N - R)$; α_{95} (A_{95}), 95% confidence cone about average direction (VGP); strike, dip, attitude of bedding, strike measured east of north, dip measured 90° to the right of the strike (where variable attitudes occur at a site, a mean value is given); J_{nrm}, J_{cl}, geometric mean intensities before and after magnetic cleaning, respectively ($\times 10^{-5}$ emu/cm^3); remag., remagnetization; NA, not applicable; N.D., not determined.

Site mean directions of magnetization for the Dekkas Andesite are shown in Figure 5 along with their associated ovals of confidence. Although the Hawkins Creek transect is a nearly homoclinal section, enough variation in attitude occurs to attempt a fold test. The beds generally strike ENE and are tilted southward between 50° and 105° (overturned). The grouping of directions for sites De2–De8 (the reversely magnetized sites) is better after correcting for the attitude of bedding, and the precision parameter k increases from 11.9 to 27.2. The fold test is significant at the 90% confidence level but not at 95% confidence using the significance test of *McElhinny* [1964]. Recently, *McFadden and Jones* [1981] have shown that McElhinny's test is incorrect and too stringent. However, McFadden and Jones' test requires rigid conditions not met by these data. No available numeric test seems entirely correct for evaluating the significance of the fold test for these sites (see also *Fisher et al.* [1987, p. 211]), but the data (Figure 5) strongly suggest that the remanence was acquired prior to folding. All sites except one are of reversed polarity, in accord with their having been magnetized during the Kiaman Reversed-Polarity Superchron [*Irving and Parry*, 1963] which lasted from late Carboniferous throughout most of Permian time. Only the site near the stratigraphic top of the Dekkas Andesite at this transect (De9) is of normal polarity, again in accordance with known polarity history which has the Kiaman ending just below the Permian-Triassic boundary [*Irving and Parry*, 1963; *Heller et al.*, 1988]; however, the possibility of remagnetization of this and site De1 will be considered below.

Nosoni Creek Transect

Permian strata at the Nosoni Creek transect were sampled at four localities within the Nosoni Formation east of the McCloud River Arm and one locality near the base of the Pit Formation. The Nosoni Formation contains mudstones, siltstones, lava flows, and abundant black shale intercalated with pyroclastic rocks similar to those that dominate the overlying Dekkas Andesite. The stratigraphically lowest of these Nosoni Formation sites (No2, Table 1) was overprinted with a postfolding, normal-polarity, northeast directed magnetization direction. All samples moved away from the overprint direction with progressive af or thermal demagnetization, and many contained only a single component of magnetization at the higher demagnetization steps. However, no consistent site-mean direction was found, indicating that little primary magnetization was retained by these samples. After correcting for tilt, 17 of the 19 samples collected had negative inclinations, indicating that they were initially of reversed polarity. In most, their "cleaned" remanence was directed toward the east or southeast.

Stratigraphically higher, locality No3 provided enigmatic results. The rock at this locality is a well-indurated, columnar-jointed pyroclastic unit that is bright red in color. The af demagnetization proved to be an ineffective cleaning technique although thermal demagnetization resulted in a good grouping of sample directions. However, the resulting mean direction (negative, eastward directed inclination, see Table 1) is unlike any yet found in the terrane with the possible exception of the general indication of a similar direction in underlying site No2. This grouping may be fortuitous and result from incomplete cleaning as appears to be the case at the underlying locality, or because the structurally corrected

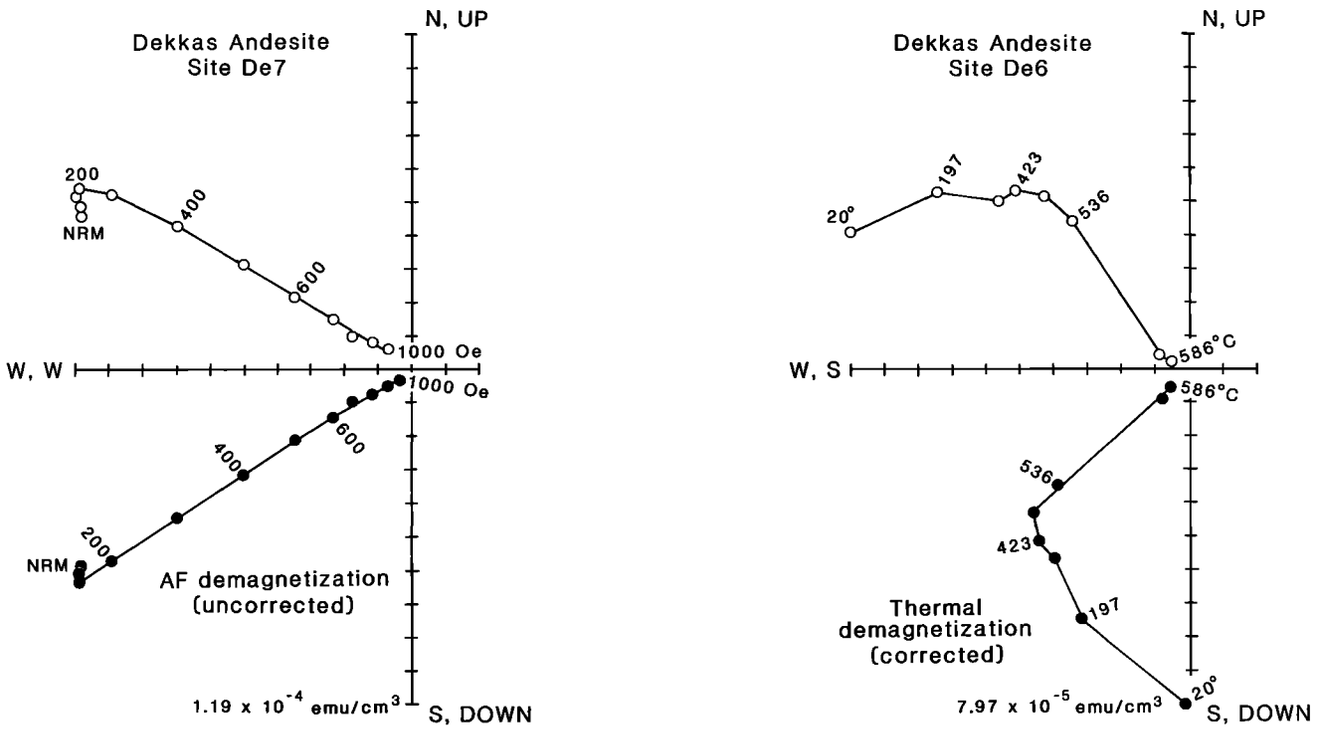


Fig. 4. Orthogonal projection of remanence vector endpoints during progressive af and thermal demagnetization of representative samples from the Dekkas Andesite. Solid symbols, projections into the horizontal plane; open symbols, projections into the vertical plane.

results from the other Permian rocks sampled, local rotation may have occurred.

Samples taken of limestone immediately southeast of the Nosoni Creek bridge (locality No4) proved to be severely overprinted with a normal polarity component. A stable remanence could not be isolated with either af or thermal demagnetization.

Samples were obtained from tuffaceous sediments at the Nosoni-Dekkas Andesite contact as mapped by *Coogan* [1960]. The NRM directions of samples from this locality (De1) were badly scattered, the intensity of magnetization was fairly weak (log mean = 2.83×10^{-6} emu/cm^3), and directional changes using either demagnetization procedure were very erratic. For these reasons, the remaining specimens from this locality were carefully polished to remove any traces of metal that may have adhered to the cores during drilling or sawing. Effects of this procedure were quite dramatic. Repeated natural remanent magnetization (NRM) measurements were less scattered than the original measurements, the remeasured intensity was much weaker (3.40×10^{-7} emu/cm^3), and af demagnetization proved effective in recovering a stable remanence direction for all samples from this locality (Figure 5). Contamination of the magnetic signal of a sample during coring or specimen preparation is not unknown [e.g., *Champion et al.*, 1984], but such effects had not been previously detected in other rocks from the study area. However, with this demonstration that some rocks in the area appear to be highly vulnerable to spurious magnetizations, all weakly magnetized samples were routinely examined and polished before proceeding with magnetic measurements.

The Dekkas Andesite is poorly exposed along the road following Nosoni Creek and no localities other than that at the Nosoni-Dekkas contact (De1) were sampled for paleomagnetic study. Locality (Pi2) was sampled in the Pit Formation near the base of the unit as mapped by *Coogan* [1960]. Siliceous shales and siltstones which are typical of the Pit are exposed at this locality although numerous chert layers are also present. It is from one of these tuffaceous chert beds that Permian radiolarians [*Silberling and Jones*, 1982] were recovered. Paleomagnetic samples from the siltstone possess a normal polarity, and a mean direction calculated for the site (Figure 5) is antipodal to the reversed-polarity sites along Hawkins Creek.

Overprint Direction

It is an unfortunate circumstance of the structural attitude at many sites in the Eastern Klamath terrane that a primary normal-polarity magnetization antipodal to the primary reversed-polarity Dekkas Andesite magnetizations will have an in situ direction near to that produced by the axial dipole field. Proximity of a normal-polarity in situ direction to the Holocene field direction will, therefore, be of little value in assessing whether a remanence is primary or secondary. It is apparent, however, that six of the seven Dekkas reversed-polarity sites at the Hawkins Creek transect have been affected by postfolding, normal-polarity overprint and that the direction of the overprinting field is located somewhere in the northeastern quadrant of the lower hemisphere of Figure 6. The remanence direction at site De6 was not affected by any overprint, so this site is not represented in Figure 6. In order to help evaluate the significance of any normal-polarity direction that might be encountered in other rocks, we used the principal component method of *Kirschvink* [1980] to determine accurately the direction of the overprinting field. Planes, unconstrained in space, were fitted to those sets of demagnetization vector endpoints for individual samples that appeared to fall on great circle paths

on an equal-area projection. To ensure that only the best defined planes were used in the final analyses, we used only those (1) that were determined from at least five vector endpoints, (2) that had a maximum angular deviation of the endpoints from the fitted plane less than 25°, and (3) for which the arc length in the plane between the endpoints was at least 40°. Bingham statistics were calculated for the 28 planes that met these minimum criteria, and these planes converged on a direction with an inclination of 68.9° and a declination of 34.0°. Semiaxes of the 95% confidence ellipse about this average direction are 4.9° and 9.4°.

A strong, postfolding, northeast directed overprint affecting the strata at localities No2 and No4 at the Nosoni Creek transect appears to be similar to that seen in the Permian strata at Hawkins Creek. Sixteen samples from these two localities yield planes that satisfy the above selection criteria. Combining these planes solutions with the 28 from the Hawkins Creek transect provide an accurate estimate of the overprinting field affecting both transects. Results for these 44 samples converge on a direction with I = 60.8°, D = 28.9°, and semiaxes of the Bingham confidence ellipse equal to 4.6° and 6.2° (Figure 6). As shown below, this remagnetization probably occurred during Middle or Late Jurassic time.

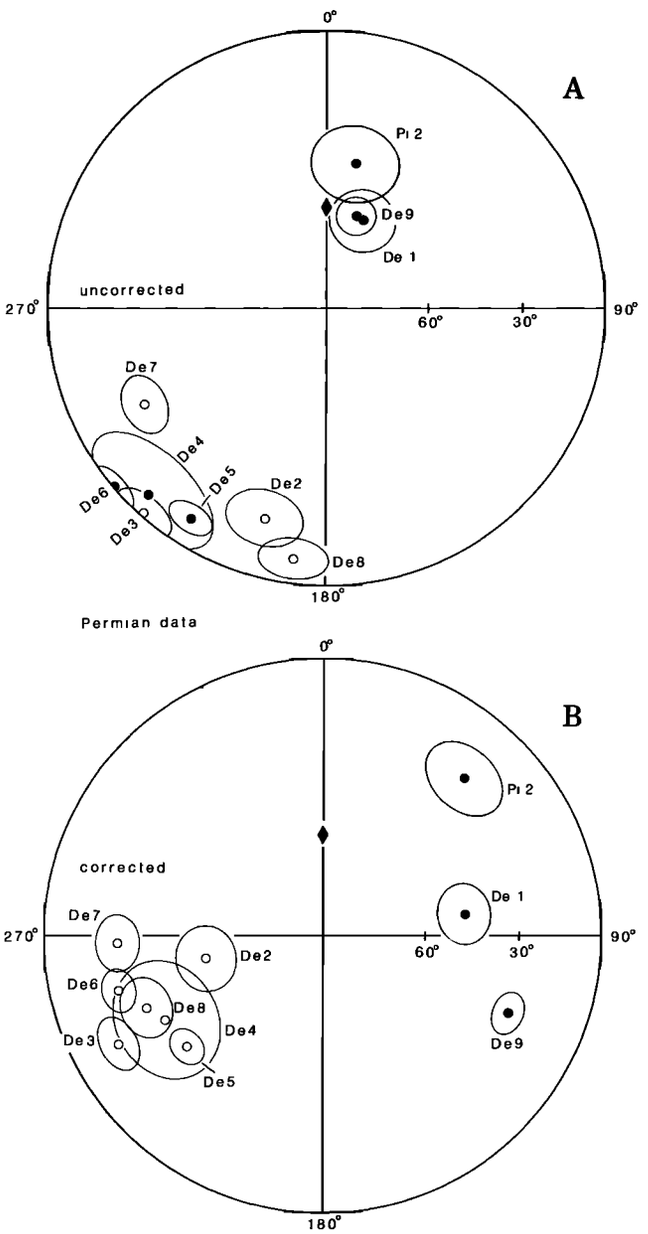


Fig. 5. Average remanent magnetization directions after magnetic cleaning for sites within the Dekkas Andesite and the Permian part of the Pit Formation. Ovals, 95% confidence limits about mean directions (*a*) uncorrected and (*b*) corrected for attitude of bedding. Solid symbols, directions on lower hemisphere of equal-area projection; open symbols, directions on upper hemisphere. Diamond, average geocentric, axial-dipole field.

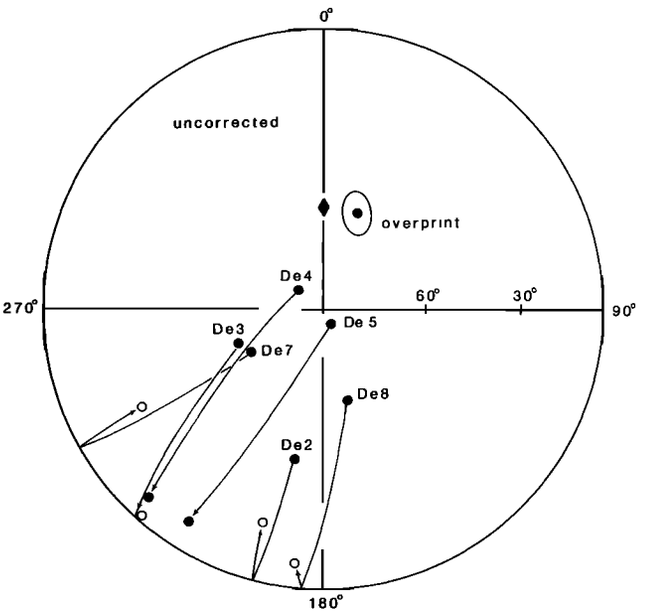


Fig. 6. Average remanent magnetization directions before and after magnetic cleaning for six reversed-polarity sites within the Dekkas Andesite at Hawkins Creek. Arrows point toward cleaned direction. Directions shown are uncorrected for bedding attitude. Overprint direction and confidence oval calculated using principal component method of *Kirschvink* [1980] on data for individual samples from Permian sites at both transects. Symbols as for Figure 5.

All samples from locality No3 at Nosoni Creek displayed similar demagnetization paths for heating steps between 300° and 550°C, and only a poorly defined planes intersection could be obtained using principal component analysis. The estimated direction (uncorrected) has an I = 20.3° and D = 19.2° which is within 14° of the mean NRM direction (Table 1) for the unit. This remagnetization is distinct from the main overprint direction given above, and its significance is problematic.

Coincidence of a direction with the strong overprinting field that has affected both transects should render a direction highly suspect, whereas coincidence with the Holocene field does not, by itself, prove remagnetization. A comparison with the overprint direction, therefore, is a necessary test of reliability for normal-polarity sites within Permian and younger strata. Accordingly, we find that the in situ directions of both normal-polarity Dekkas Andesite localities (De1 and De9) and the mean overprint do not differ statistically, indicating that the strata at those sites have been remagnetized. The site-mean directions for both will be excluded from further calculations. The direction of the normal-polarity site within the "Permian age" Pit Formation at the Nosoni Creek transect (Pi2) is statistically distinct,

however (Figure 5), and is considered to represent a primary magnetization. When the magnetization direction for site Pi2 is combined with those for reversed-polarity sites De2–De8 (Table 1), we find that k for the sample population increases from 9.2 to 23.0 after correction for bedding attitude. This fold test is significant at the 95% confidence level [*McElhinny*, 1964].

Previous Studies in the Nosoni Formation

The strata at our locality De1 were sampled by *Fagin and Gose* [1983], but the mean directions that they report differ significantly from those which we calculate (Table 1). There is an angular difference of 22° between the uncorrected directions and 40° between the corrected means. The cause of this discrepancy is uncertain, but the susceptibility of the rock to spurious magnetic components may be a contributing factor. *Fagin and Gose* [1983] rejected the results from this locality in their final analysis because the calculated pole position differed from those for their other sites and because both polarities were found in contrast to most Permian rocks. Only normal polarities were found by us; we agree, however, that this site has been remagnetized as described above.

Our site No3 was sampled also by *Renne et al.* [1988] (their site PR14), who obtained the same in situ result that we report (Figures 3 and 10 [*Renne et al.*, 1988; P. R. Renne, oral communication, 1985]). There is, however, a disagreement as to the structural attitude of the site. The attitude that we report (strike 357°E, dip 20°NE) is concordant with the regional structure (also see *Coogan* [1960]) and approximately perpendicular to the columnar jointing, and the magnetic inclination with respect to the bedding plane that we measure (−44°) is similar to other Permian inclinations at both transects (Table 1). The attitude (strike 346°E, dip 30°SW) reported by *Renne et al.* [1988] is nearly perpendicular to the regional structure, and the magnetic inclination with respect to their bedding plane (+5°) is anomalous. Those authors consider the direction recorded by this site to be a result of unremoved secondary components, although it is not clear that this is indeed the case. If *Renne et al.*'s [1988] reported structural attitude is incorrect, as we believe, it probably will not affect their major conclusions except that their prefolding direction determined by the planes analysis will be less well constrained.

Renne et al. [1988] obtained paleomagnetic samples from seven additional sites within the Nosoni Formation, two of which were considered to have been remagnetized during Quaternary time. Of the six sites that they consider to yield meaningful results (including our No3, their PR14), three are within a single cooling unit, an ignimbrite referred to as the "Skookum Tuff." The data appear to pass a fold test, and the resulting paleopoles are considered to be representative of cratonic North America. The fold test is not very compelling, however, because of the limited sampling and because the test depends critically on the results from one of the sites (PR7) within the ignimbrite. Except for its value in the fold test, *Renne et al.* [1988] are mistrustful of the results from this site because it has a large structural dip and, consequently, data from site PR7 are not used in their analysis of remagnetization planes. Furthermore, the fact that only three separate tuffs containing the presumed primary remanence direction were sampled (not including site PR14; see their Table 1) makes it very unlikely that secular variation was averaged in their data set. Their conclusion that the resulting paleopole is representative of the Permian geomagnetic field, therefore, is questionable. *Renne et al.* [1988] note that little or no secular variation is evident in their results but attribute this lack to the possibility that secular variation is at a minimum during long, stable polarity chrons such as the Kiaman. Our data, on the other hand, show considerable between-site dispersion (k = 23) that is consistent with normal secular variation during this period. *Renne et al.* [1988], however, do appear to have isolated a magnetization direction that differs from most others reported from the province.

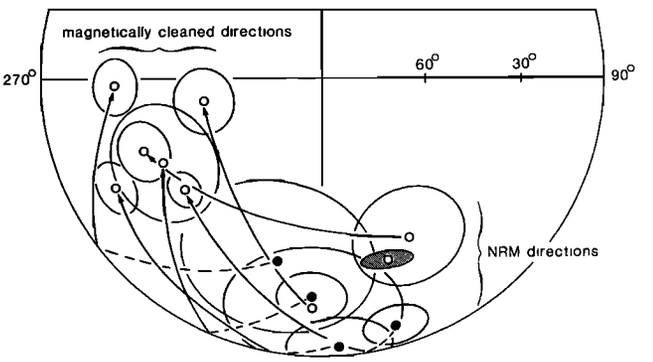


Fig. 7. Structurally corrected NRM and magnetically cleaned site-mean directions for Dekkas Andesite sites shown in Figure 5. Shaded ellipse, prefolding direction determined by *Renne et al.* [1988]. Symbols as for Figure 5.

Summary of Results From Permian Strata

Reversed-polarity results from sites within the Dekkas Andesite at Hawkins Creek are considered to represent a primary magnetization because of their behavior during af and thermal demagnetization (Figure 4) and because of the improvement in grouping of site-mean directions after correction for tectonic tilt (Figure 5). The normal-polarity site near the base of the Pit Formation (Pi2) at Nosoni Creek also is considered to be primary because it is antipodal to the reversed-polarity Hawkins Creek sites and because it was unaffected by the strong overprint noted in the underlying rocks at both transects. Only the uppermost site within the Dekkas Andesite at Hawkins Creek (De9) has been remagnetized, whereas all of our sites within the Nosoni Formation and Dekkas Andesite at Nosoni Creek have been severely overprinted to the extent that a primary magnetization could not be recovered. The cause of the pervasive overprinting at Nosoni Creek is unknown, unless a downward increase in metamorphic grade and degree of recrystallization that *Renne et al.* [1988] and *Renne and Scott* [1988] report for strata in that area is a factor. However, there is no substantial difference in metamorphic grade between the two transects, as both appear to have been subjected to prehnite-pumpellyite to low-greenshist metamorphism [*Curtis*, 1983; *Renne and Scott*, 1988; *Miller*, 1989].

Although *Renne et al.* [1988] have isolated a single component of magnetization for their sites at Nosoni Creek, the significance of their result is difficult to assess. We note, for example, that the structurally corrected, "common axle" direction that they report is located in the same general area as are the structurally corrected, site-mean NRM directions

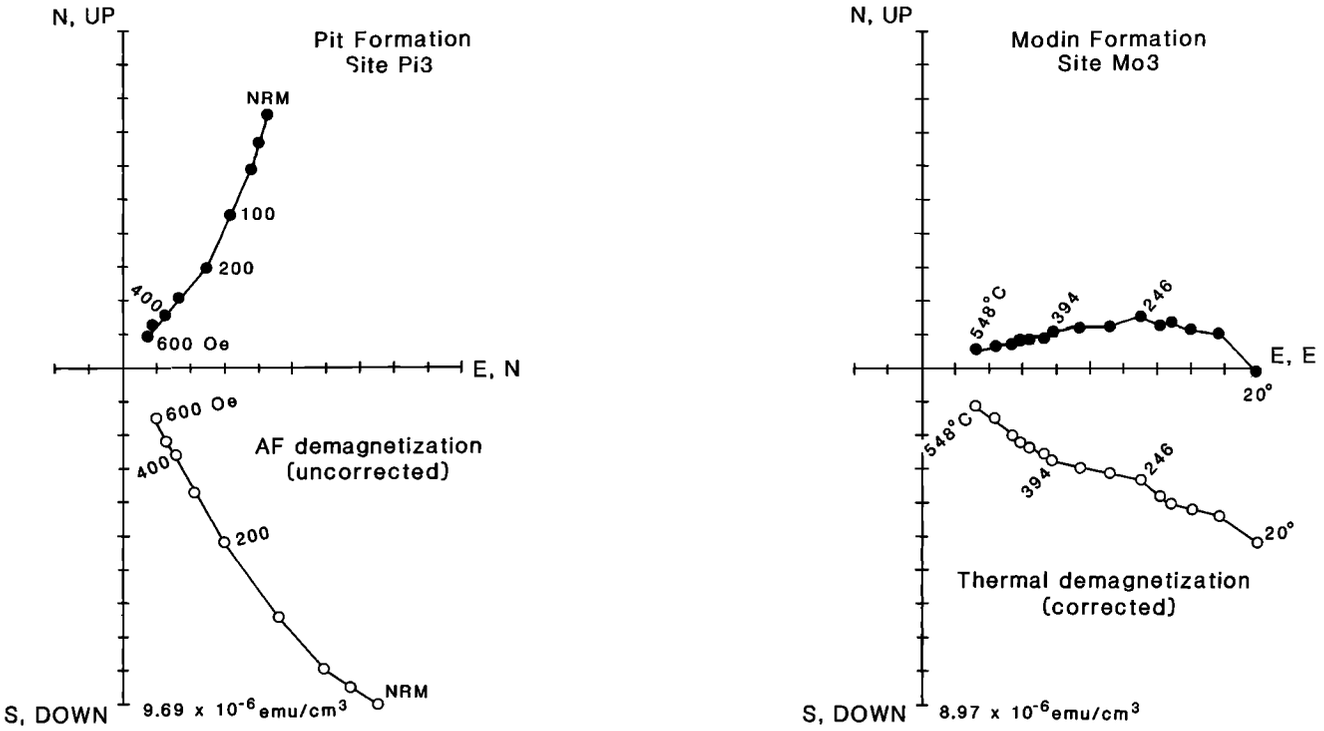


Fig. 8. Orthogonal projection of remanence vector endpoints during progressive af and thermal demagnetization of representative samples from Triassic strata. Symbols as for Figure 4.

for our Hawkins Creek sites (Figure 7). This crude agreement in direction could be fortuitous only but suggests to us that their mean direction is in fact the sum of superimposed components of magnetization whose coercivity and blocking temperature spectra are similar. If so, *Renne et al.*'s [1988] result has no more significance than our results for sites No2 and No3 described above. Because the magnetization of the Nosoni Formation seems to be very complex, and because Renne et al.'s sampling was very limited (small N) and confined to a restricted geographical area, we believe that a primary magnetization for these strata has yet to be demonstrated.

TRIASSIC STRATA

We obtained 61 samples from six localities in the Pit Formation, which along Hawkins Creek consists primarily of dark shales and siltstones with abundant tuff layers. The overlying Modin Formation along the Hawkins Creek transect consists of three members which were described fully by *Sanborn* [1960]. Our paleomagnetic sampling was confined to the middle member which consists of massive, fossiliferous limestone and calcareous sandstone. We obtained a total of 29 cores from two limestone localities and one sandstone unit near the top of the member. Strata that are definitely known to be Triassic in age were not sampled at the Nosoni Creek transect owing to poor exposures.

The rocks at each of the Modin Formation localities proved to carry a stable remanence (Figure 8), and all were of normal polarity. Unfortunately, only the two lowermost sample localities in the Pit Formation provided usable results. All of the samples from the lowest site (Pi3) were of normal polarity and had good magnetic stability (Figure 8). Four of the samples from the second site (Pi4) originally were of reversed polarity, but these became magnetically unstable before stable endpoints could be reached with either af or thermal demagnetization. The remaining samples showed no tendency toward reversed-polarity directions and are considered to have a primary normal polarity.

Site mean directions of magnetization for the rocks of Triassic age (Pit and Modin formations) are shown in Figure 9 along with their associated circles of confidence. Because some of the uncorrected directions are close to the direction that would be produced at the site by a geocentric axial dipole and to the overprint direction shown in Figure 6, the possibility of remagnetization has to be considered. All sites have magnetization directions that differ from the overprint at confidence levels greater than 70%, and the mean for site Mo3 is distinct at 100% confidence. We assume, on the basis of this test, that these sites may have been affected little, if at all, by the overprinting field. We also note that although the similarity of bedding attitudes at each of the five sites precludes a conclusive fold test, the grouping of site means is marginally better after applying the tilt correction. Where differences in bedding attitude occur at an individual site, correcting for tilt also improves the grouping of specimen directions (Table 1), but again, the differences are slight. Because of these indications and the fact that the remanence of the samples appears to be stable (Figure 8), we will consider, for the present, that the directions obtained are probably primary. The fact that the directions are close to what would be expected based on our results from the Permian strata adds some measure of confidence to this conclusion.

Previous Studies in the Modin Formation

A 2-m-thick limestone bed at our locality Mo1 (Table 1) was sampled by *Fagin and Gose* [1983]. There is an angular difference of 13° between the uncorrected direction that they

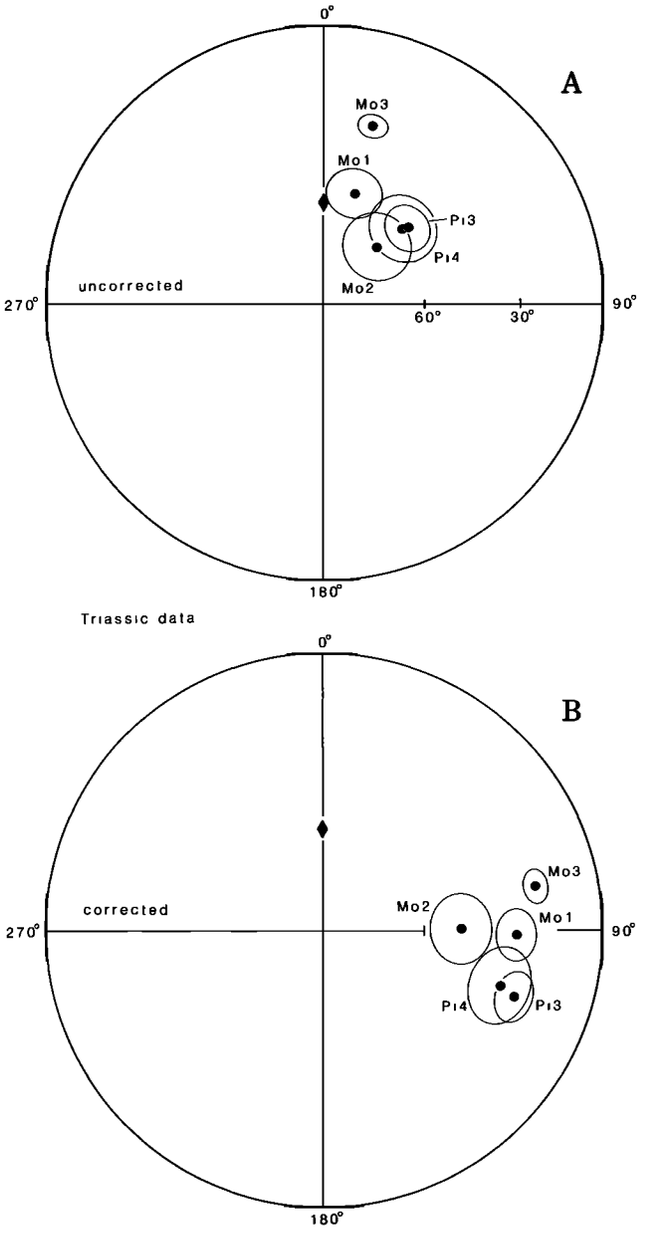


Fig. 9. Average remanent magnetization directions after magnetic cleaning for sites of Triassic age (*a*) uncorrected and (*b*) corrected for attitude of bedding. Symbols as for Figure 5.

obtained and the one we report in Table 1, and the corrected means differ by 20°. Both sets of mean directions are statistically different at the 95% confidence level using the method of *McFadden and Jones* [1981]. The discrepancy probably arises because the sampling by *Fagin and Gose* [1983] was confined to a single limestone bed, whereas our sampling spanned several such beds over a stratigraphic thickness of approximately 40 m. Three samples we obtained from the same thick limestone bed yield magnetization directions that are very close to the mean reported by *Fagin and Gose* [1983], and we conclude that our sampling scheme better averages the effects of geomagnetic secular variation.

JURASSIC STRATA

The upper member of the Modin Formation is very thin or absent along the Hawkins Creek transect, and the predominantly limestone middle member is overlain unconformably by the Arvison Formation of *Sanborn* [1960]. The name Arvison Formation was proposed by *Sanborn* [1960] for a series of dominantly pyroclastic rocks that occur between the Modin and Potem formations. In the type area, the unit consists primarily of volcanic breccia and conglomerate, with minor lava flows and tuffaceous sandstone. Paleomagnetic samples were obtained from two localities (Ar4 and Ar6) of gray-green, impure volcanic sandstone at the Hawkins Creek transect.

Four localities within the sequence of Jurassic rocks were sampled near the southeast end of the Nosoni Creek transect. Sites Ar3 and Ar5 (Table 1) were sampled in what was originally mapped as Potem Formation by *Diller* [1906]. Site Ar3 near Potem Creek is near the base of the unit as it was mapped, and site Ar5 is stratigraphically higher along the road toward the top of Bluejay Mountain. However, both sites are in accumulations of impure volcanic sandstone that look similar to the Arvison Formation as mapped by *Sanborn* [1960] at the Hawkins Creek transect. Furthermore, rock that is closer in appearance to the Potem Formation at the Hawkins Creek transect occurs just above site Ar5, as do exposures of the Bagley Andesite. We believe that these two sample localities are in rocks equivalent to the Arvison Formation of *Sanborn* [1960].

Site Ar1 was sampled south of the two localities just described and east of the Pit River in rock that *Diller* [1906] mapped as the Bagley Andesite. The rock at this locality is a coarse volcanic breccia that also is probably equivalent to the Arvison and low in the Jurassic section. Site Ar2 was sampled approximately 2.5 km southeast of site Ar1, where a series of thin, approximately 0.5- to 1-m-thick beds of coarse-grained, impure volcanic sandstone is exposed. A red-weathering surface is commonly developed on the rocks at this locality. As with the other three localities just described, these rocks are probably correlative with the Arvison Formation of *Sanborn* [1960].

Conformably overlying the Arvison is the Potem Formation and the intercalated Bagley Andesite. The Potem Formation consists primarily of argillite and tuffaceous sandstone, with minor beds of conglomerate, pyroclastic rocks, and limestone. The argillite in the Potem is dark gray to black where fresh but weathers occasionally to an intense reddish or purplish color. Both localities that we sampled near the Pit River on the south side of Oak Mountain have this coloration. The Bagley Andesite was not sampled for this study because of poor exposures and common lack of adequate structural control. The lack of structural attitudes in the Bagley may be explained by new data [*Renne et al.*, 1986; *Renne and Scott*, 1988] indicating that the Bagley is intrusive, at least in part. These authors also report K-Ar determinations on plagioclase separates that yield ages of 166 ± 2 Ma and 168 ± 3 Ma for two different Bagley Andesite stocks. These data help to confirm, in part, the previous Early and Middle Jurassic age assignments [*Sanborn*, 1960] for these two formations.

A stable remanence could be isolated readily in the Jurassic rocks with either af or thermal demagnetization (Figure 10). Maximum blocking temperatures for samples from the Arvison were clustered near 500°C and were between 540° and 560°C for samples from the Potem. The blocking temperatures for the Potem samples indicate that the remanence of these rocks is carried by magnetite despite their color. One specimen was acid leached for a total of 90 hours to remove the red coloration and was then progres-

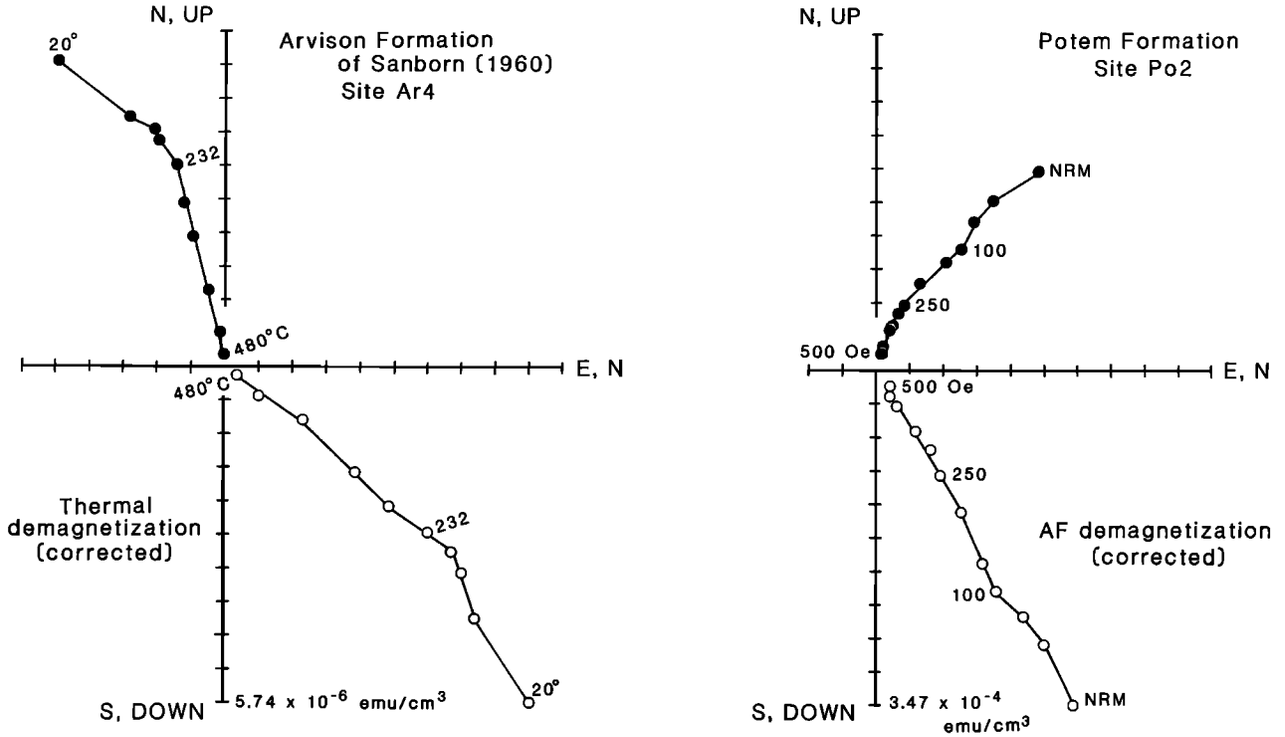


Fig. 10. Orthogonal projection of remanence vector endpoints during progressive af and thermal demagnetization of representative samples from Jurassic strata. Symbols as for Figure 4.

sively af demagnetized. This sample yielded a direction that remained within the main group of directions of all samples from the site demonstrating that little, if any, remanence is carried by the red pigment.

Site-mean directions for rocks of Jurassic age at both transects are shown in Figure 11. Strata at all sites were found to possess a normal polarity, and all except Ar5 have in situ directions that are statistically distinct from the mean overprint direction. The direction at site Ar5 differs from the overprint at 90% confidence. Correcting for bedding attitude yields three distinct groups of magnetization directions that seem to be a function of stratigraphic position within the sequence. Site Ar1, thought to be the oldest site sampled, has a shallow inclination and eastward declination, which indicates that the direction could be intermediate between that recorded by the Permian and Triassic rocks and by later Jurassic strata. However, because other rocks that may have recorded the same magnetization direction as that at site Ar1 were not sampled and because the true stratigraphic position of this site is uncertain, any conclusions based on this result should be considered tentative.

The younger two directions shown in Figure 11*b* are based on results from three or more sites. Although the ages of the Arvison and Potem have been thought to differ substantially [*Sanborn*, 1960], the youngest site sampled at the Nosoni Creek transect (Ar5) yields a direction that is similar to that recorded by the two Potem sites and Arvison site Ar6 at the Hawkins Creek transect. The age of the Arvison is based on fossils from a single locality whose stratigraphic position in the formation as a whole is not known, and the age of the Potem is similarly poorly constrained because of the lack of a clearly diagnostic fossil fauna [*Sanborn*, 1960]. Thus it is possible that the sampled localities do not differ greatly in age. The precision parameter k for this younger group of site-mean directions (Ar5, Ar6, Po1, Po2) increases from 19 to 103 when corrected for bedding attitude and is significant at 95% confidence. Stratigraphically lower, sites Ar2 and Ar3 at the Nosoni Creek transect yield directions that are close to that recorded by site Ar4 at the Hawkins Creek

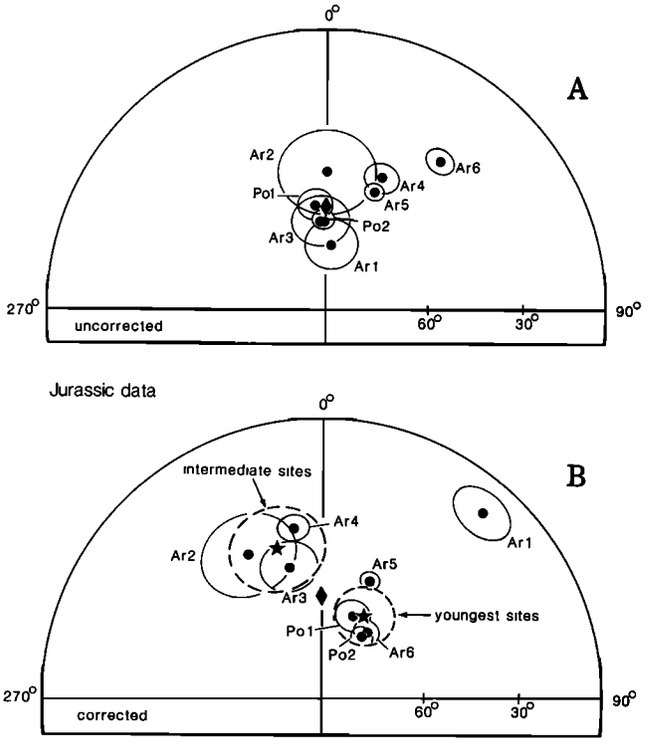


Fig. 11. Average remanent magnetization directions after magnetic cleaning for sites of Jurassic age (*a*) uncorrected and (*b*) corrected for attitude of bedding. Stars, mean directions calculated for stratigraphically youngest and intermediate sites along with their associated 95% confidence limits (dashed). Symbols as for Figure 5.

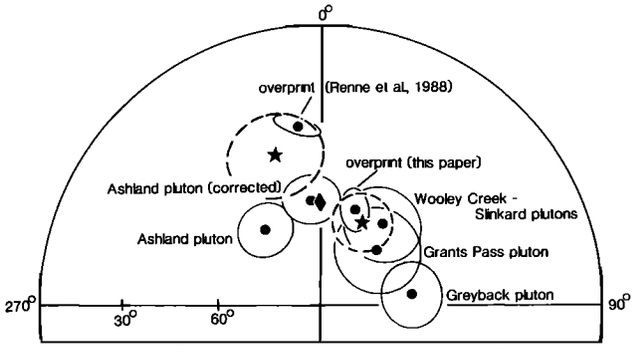


Fig. 12. Mean in situ remanent magnetization directions after magnetic cleaning for Jurassic plutons of the Klamath Mountains [*Schultz*, 1983]. Corrected direction for Ashland pluton also shown, as are mean intermediate and youngest site directions (from Figure 11). Postfolding overprint directions shown are those affecting rocks at both our transects (see text) and that affecting the sites sampled by *Renne et al.* [1988]. Symbols as for Figure 5.

transect. For these three older sites, k increases from 41 to 83, which is significant at 75% confidence. The improvement in grouping of site-mean directions after correcting for tilt of strata indicates that a prefolding magnetization has been preserved in both. That two (or three) considerably different magnetization directions could be recorded by the Jurassic rocks should not be entirely unexpected because this was a period of rapid apparent pole wander (APW) [*May and Butler*, 1986].

Related Studies in Jurassic Rocks

Evidence for the younger two directional groups described above is provided by other paleomagnetic studies in the Klamath Mountains province. Paleomagnetic samples were obtained from 28 sites within five calc-alkaline plutons in the composite Western Paleozoic and Triassic terrane in the northern part of the province by *Schultz* [1983], who found that the remanent magnetization of the plutons was stable and primary. Interpretation of paleomagnetic data from plutons is inherently difficult both because of a wide range of magnetic properties and because they usually lack the structural control necessary to detect postemplacement tilting. Despite these difficulties, the large number of sites sampled has yielded remanence directions for these plutons that, in general, seem mutually consistent. Site-mean directions for these plutons, uncorrected for any postmagnetization tilting, are shown in Figure 12. A corrected direction for the Ashland pluton is shown also, derived from the attitude of the overlying Cretaceous sedimentary Hornbrook Formation. *Schultz* [1983] combined the paleomagnetic results from the Wooley Creek and Slinkard plutons because of independent evidence [*Barnes et al.*, 1986*a*] that they probably represent different levels of the same intrusion. Although there is evidence that more of the plutons than just the Ashland may have undergone postemplacement tilting [*Barnes et al.*, 1986*b*], actual amounts and timing are not closely constrained, and a thorough evaluation is beyond the scope of this paper.

As can be seen in Figure 12, there is excellent agreement between our results for the youngest Jurassic strata and those from the Grants Pass and Wooley Creek-Slinkard plutons. Note, also, the close correspondence between these directions and that of the overprint affecting the Permian rocks at the Hawkins Creek transect and the Nosoni Creek transect east of the McCloud River Arm. The site-mean direction for the Ashland pluton may be related to our intermediate group of directions or be transitional between the two groups. An overprint direction determined by *Renne et al.* [1988] and considered to be Jurassic in age also seems to belong to this group of directions. The site-mean direction for the Greyback pluton does not correspond exactly to any of the directions shown in Figure 12 but is most likely akin to the youngest directional group, as would be expected because its radiometric age agrees with those determined for the Grants Pass, Wooley Creek, and Slinkard plutons (see summary by *Irwin* [1985]). Some differential rotation and/or tilting between this and the other plutons that were studied may have occurred.

Summary of Results From Jurassic Strata

Even though the sites that we sampled seem to have been unaffected by the strong overprint detected in rocks from both transects, the number of in situ directions that are close to the axial dipole field for the area may lead one to suspect a Recent remagnetization for some of the strata, especially the Potem. However, when all data are considered, the results, corrected for bedding attitude, yield a bimodal distribution of magnetization directions (Figure 11) if site Ar1 is not included for the reasons listed above. The mean directions of both groups (Table 2) are statistically distinct at the 99% confidence level [*McFadden and Jones*, 1981], and each seems to be a function of stratigraphic position within the sequence. Behavior during thermal and af demagnetization shown in Figure 9 is representative of samples within both directional groups and indicates magnetic stability. Strong fold tests, one of which is significant at the 95% confidence level, indicates that the magnetization of sites within each directional group is prefolding. Evidence for both directional groups is found at both transects and is provided also by other paleomagnetic studies in the Klamath Mountains. Therefore we have no good reason for disregarding either (or both) group(s) of directions and, for the present, will consider that the Jurassic rocks that we have sampled do retain a primary remanence, and the differences seen may, at least in part, be due to APW during this time interval and/or a complex rotational history.

MAGNETIZATION OF THE MCCLOUD LIMESTONE

We did not sample the McCloud Limestone during the earlier phases of our study because the unit commonly is dolomitized, or otherwise altered to some degree. Much of

Table 2. Mean Paleomagnetic Poles for Strata of the Eastern Klamath Terrane

	N	R	k	A_{95}	PLA	PLO
Jurassic sites (uppermost)	4	3.9564	69	11.2°	69.7°	306.1°
Jurassic sites (intermediate)	3	2.9722	72	14.7°	69.5°	105.8°
Triassic sites	5	4.8789	33	13.5°	8.4°	310.6°
Permian sites	8	7.7151	24	11.4°	30.6°	329.2°

All data corrected for attitude of bedding. PLA, PLO, north latitude and east longitude, respectively, of paleomagnetic pole. See Table 1 for additional explanation.

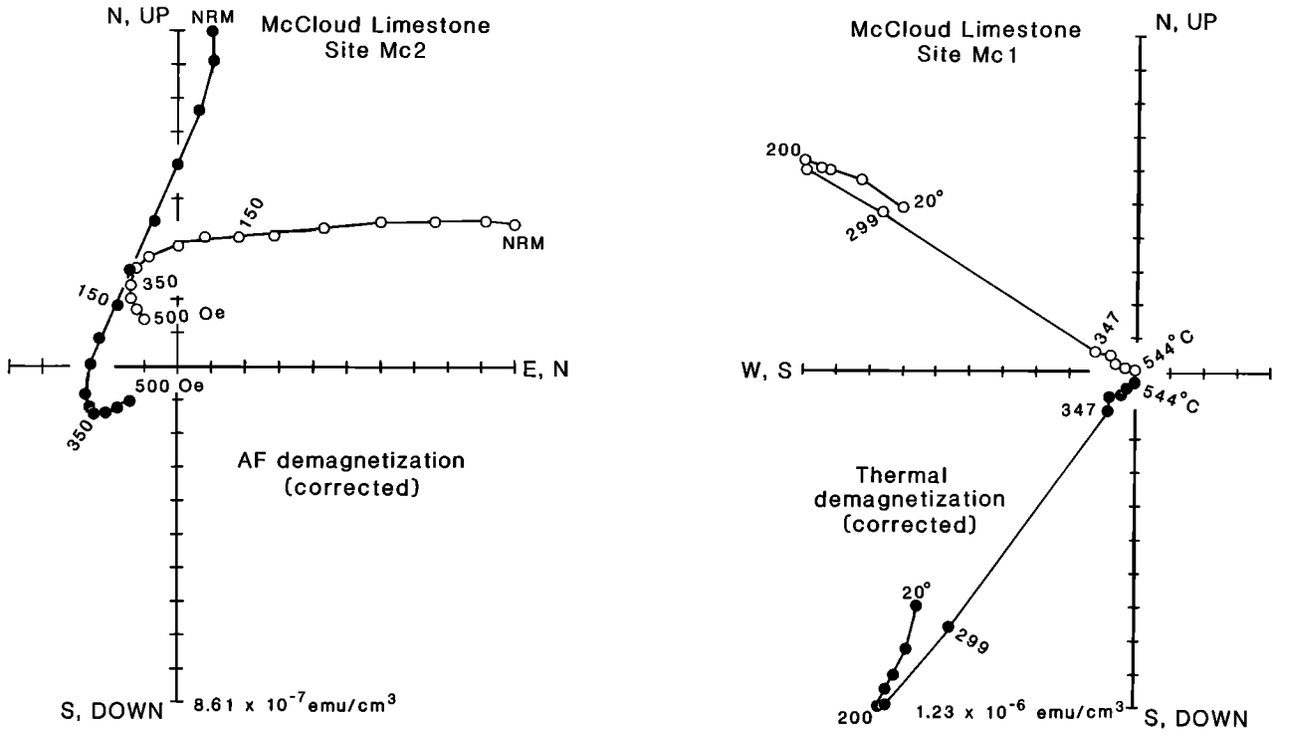


Fig. 13. Orthogonal projection of remanence vector endpoints during progressive af and thermal demagnetization of representative samples from the McCloud Limestone at the Nosoni Creek transect. Symbols as for Figure 4.

the McCloud is found in close association with mafic quartz diorite and other intrusive rocks that have caused extensive recrystallization of the limestone, and in some areas, it is a host for contact metasomatic magnetite deposits [*Albers and Robertson*, 1961]. For these reasons, remagnetization of the unit seemed likely. Nonetheless, *Fagin and Gose* [1983] have reported a reversed-polarity direction for the limestone along the McCloud River Arm of Shasta Lake that they considered to be primary because its polarity agreed with what would have been expected based on the age of the unit. The structurally corrected direction that they reported agreed well with the directions that we had obtained from the reversed-polarity Dekkas Andesite localities at the Hawkins Creek transect (Figure 5*b*) and seemed indicative of a Permian magnetization. Because *Fagin and Gose* [1983] did not address the possibility of remagnetization, we sampled additional localities to test whether the magnetization of the McCloud Limestone is primary or secondary. We sampled two localities within the limestone, a locality within the Nosoni Formation that is in close proximity to the intrusion, and two localities within the quartz diorite itself.

The localities sampled are near the northwest end of our Nosoni Creek transect (Figure 2). One outcrop of the McCloud Limestone (Mc1) was sampled at the same locality reported by *Fagin and Gose* [1983]. The diorite intrusion is exposed at both ends of the limestone outcrop at this locality but is not suitable for paleomagnetic sampling. The intrusion is not exposed at the second limestone locality (Mc2), which is approximately 0.3 km north of Mc1, but certainly occurs within a few meters of the outcrop [see *Coogan*, 1960, Figure 2]. The mafic quartz diorite was sampled at two localities, one of which is approximately 0.3 km south of McCloud Limestone locality Mc1. The diorite here (mqd3) forms a dike that is nearly perpendicular to the bedding of the McCloud at the same locality. The second locality (mqd2) is approximately 0.6 km to the southeast where a few small dikes intrude the McCloud and more massive diorite occurs at the northeast end of the limestone exposure. However, the massive exposure is badly fractured, and a total of only five cores were obtained from both occurrences of diorite at this locality. The locality within the Nosoni Formation (No1) is approximately 2.5 km SE of our diorite localities but within about 600 m of the trend of the intrusion as mapped by *Coogan* [1960].

Results

The NRM intensities of samples from the McCloud Limestone were very weak (log mean = 4.2×10^{-7} emu/cm^3) and most possessed a postfolding, normal-polarity, northwest directed magnetization direction. In many cases, both af and thermal demagnetization were effective in removing this normal-polarity overprint (Figure 13). We progressively af demagnetized one specimen from each core and later selected a companion specimen from the same core for thermal demagnetization if the first treatment was not effective. By this procedure, a stable magnetization was isolated in 65% of the cores collected (Table 1). A stable magnetization was most readily isolated in the samples from the outcrop sampled by *Fagin and Gose* [1983], and the mean direction we obtained (Mc1, Table 1) is statistically indistinguishable from the one that they reported. The postfolding overprint is more pronounced at the second locality, and a stable remanence could be isolated in only nine of the 18 cores collected.

Localities within the Nosoni Formation and the mafic quartz diorite also were affected by a northwest directed, normal-polarity overprint. That within the Nosoni (No1) was easily removed by af or thermal demagnetization and a peak field of 300 Oe was used to clean all samples from this locality. All samples from diorite locality mqd3 could be cleaned by either af or thermal demagnetization, but the

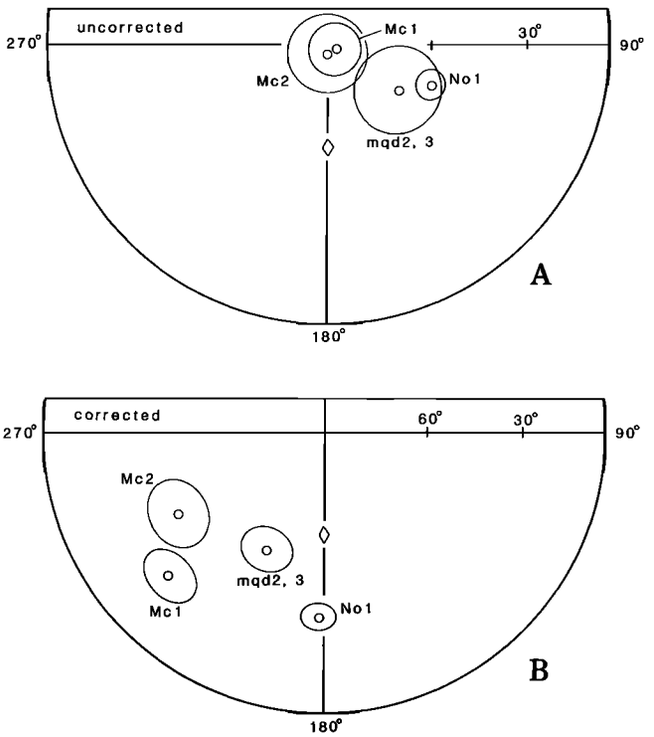


Fig. 14. Average remanent magnetization directions after magnetic cleaning for sites near the Nosoni Creek transect that were sampled to test the possibility of remagnetization by the mafic quartz diorite (mqd). Directions shown (*a*) uncorrected and (*b*) corrected for attitude of bedding. Symbols as for Figure 5.

overprint at locality mqd2 was much stronger and could be removed in only two of the five samples collected. For this reason, sample results from the two diorite localities were combined into a single mean direction in Table 1 and Figure 14. As a first approximation, bedding measured in the McCloud Limestone at both localities was used to structurally correct the subsite results from the diorite. In doing so, we assumed that the folding of the limestone postdated the acquisition of remanence by the intrusion, although this may or may not be correct.

Nature of the Remanence

Site-mean directions of magnetization for the two McCloud Limestone localities, Nosoni site No1, and for the diorite intrusion are shown in Figure 14 along with their ovals of confidence. It is quite obvious that the grouping of directions is better before correcting for the attitude of bedding. This negative fold test shows that the McCloud and Nosoni localities contain a postfolding remanence and that the remagnetization probably was caused by the intrusion. The divergence of the McCloud and Nosoni directions after unfolding indicates that the intrusion caused some local deformation. The difference in direction before unfolding between the McCloud sites on one hand and the diorite and Nosoni sites on the other may be the result of local relative movements after the intrusion cooled but before the main episode of deformation of the province. The convergence of the two diorite subsites upon unfolding is probably a happenstance. Even though these McCloud and Nosoni sites have been remagnetized, the general agreement in polarity and magnetic inclination with the primary magnetizations in the Permian strata at Hawkins Creek is an indication that the remagnetization, and therefore the intrusion may be Permian in age.

No successful age determinations have been made on the diorite intrusion, but it has generally been thought to be related to the Pit River stock which is Permian in age [*Lanphere et al.*, 1968; *Fraticelli et al.*, 1985; *Kistler et al.*, 1985; *Albers and Bain*, 1985]. Rubidium-strontium isotopic data from samples of the Pit River stock and the diorite intrusion (R. W. Kistler, written communication, 1985) are similar and thus compatible with this assumption. The fact that the magnetic inclination recorded by the intrusion (−51.5°) is steeper than inclinations in the other Permian rocks sampled is probably due, in part, to attitudes measured in the McCloud Limestone being inappropriate for structurally correcting the diorite results. If the intrusive is comagmatic with the overlying Nosoni Formation and Dekkas Andesite, correcting the results for the general attitude of the lower part of the Nosoni might be more appropriate and would, in fact, tend to shallow the measured inclinations toward typical Permian values. Also, cooling time of this small intrusion would be unlikely to have been long enough to average geomagnetic secular variation and yield an accurate representation of the Permian field. In an unsuccessful attempt to establish a closer tie between the two units, we collected paleomagnetic samples from the Pit River stock which will be described below.

Overprint Direction West of the McCloud River Arm

Because of the strong, normal-polarity remagnetization superimposed on the stable reversed polarity carried by these units, a demagnetization planes analysis could be used to identify the overprint direction. Data from all samples whose demagnetization vector endpoints appeared to fall on great circle paths were analyzed, and those that satisfied the same selection criteria as we used for the Dekkas Andesite were used in the calculation of a mean direction. The selected planes all had arc lengths ranging from 61° to 141°, demonstrating that the planes were well determined. In addition to those samples that yielded well-defined demagnetization paths, nine samples were totally remagnetized, and lines were fitted to their demagnetization vector endpoints. An estimate of a mean direction was made by combining the lines and planes data using method 2 of *Kirschvink* [1980]. Analysis of the 28 selected lines and planes yields a mean direction with $I = 65.4°$, $D = 345.1°$, and semiaxes of the Bingham confidence ellipse equal to 6.9° and 8.7°. This mean is a realistic estimate of the direction of the remagnetizing field affecting the localities west of the McCloud River Arm and differs from that determined for other Permian sites at both transects. This overprint direction does not differ significantly from that determined for remagnetized rocks of the Great Valley sequence that onlap the Klamath Mountains at the southern end of the province according to *Mankinen and Irwin* [1982], who consider the remagnetization of the Great Valley sequence strata to be no younger than latest Cretaceous or earliest Tertiary.

OTHER SAMPLE LOCALITIES IN THE EASTERN KLAMATH TERRANE

Rocks at several localities within the Eastern Klamath terrane but not part of the two transects also were sampled.

The oldest rock sampled was the granitoid Mule Mountain stock (Figure 2). The Mule Mountain stock is Devonian in age [*Albers et al.*, 1981] and represents some of the root zone of the volcanic island arc that formed the early part of the Eastern Klamath terrane. The af demagnetization was able to isolate a single magnetic component which yields an average $I = -39.7°$, $D = 241.0°$, and $\alpha_{95} = 9.8°$ (Table 1) when the sample data are structurally corrected using attitudes measured on the comagmatic Balaklala Rhyolite adjacent to the stock. If this direction is an accurate representation of the Devonian geomagnetic field, considerable southward transport of the terrane would be required. However, the coincidence of this direction with the Permian reversed-polarity directions from the Hawkins Creek and Nosoni Creek transects is a strong indication that the Mule Mountain stock has been remagnetized. In the absence of evidence to the contrary, we consider it most likely that this intrusion records a prefolding, secondary magnetization acquired during the Permian.

In an attempt to test the hypothesis that the Permian Pit River stock and the mafic quartz diorite that intrudes the McCloud Limestone form a comagmatic plutonic-volcanic suite with the Dekkas Andesite [*Irwin*, 1985], paleomagnetic samples were obtained from a mafic phase of the stock south of Shasta Lake (Figure 2). Unfortunately, this locality proved to be strongly overprinted with a northeast directed, normal polarity. Neither af nor thermal demagnetization was effective in removing the overprint. Commonly, thermal demagnetization to temperatures above 450° or 500°C was necessary before any change from the overprint direction occurred, and no primary remanence could be isolated. In order to establish the direction of the remagnetizing field, least squares lines, unconstrained in space, were fitted to the demagnetization vector endpoints between temperatures of 100° and either 450° or 500°C. Line data for 12 samples were averaged, and these yielded a mean in situ direction with $I = 59.7°$, $D = 28.0°$, and $\alpha_{95} = 5.8°$. This direction is identical to the overprint affecting the Permian rocks at both our transects to the north.

The mafic quartz diorite was sampled where it intrudes the McCloud Limestone approximately 1.1 km southeast of the Pit River stock locality. The diorite also has been overprinted, but the remagnetization is not as complete as in the Pit River stock. Although the overprint could not be completely removed, directional changes during demagnetization indicate a primary reversed magnetization. Planes were fitted to the high thermal (>380°C) and high af (>200 Oe) demagnetization vector endpoints for 11 samples; these converged on an in situ direction with $I = -61.2°$, $D = 193.7°$, and semiaxes of the Bingham confidence ellipse equal to 10.5° and 14.1°. Correcting for the attitude of the overlying Dekkas Andesite yields a direction ($I = -22.0°$, $D = 244.1°$) that is similar to that of the diorite at the Nosoni Creek transect. These results indicate that contemporaneity of the diorite in both areas is likely.

One exposure of quartz keratophyre from the lower part of the Pit Formation [*Albers and Robertson*, 1961] was sampled along California State Highway 299 northeast of Redding (Figure 2). NRM intensities of the samples at this locality (Pi1, Table 1) were very weak (log mean = 2.1×10^{-7} emu/cm^3), and directions were badly scattered. The samples proved to be completely unstable to both af and thermal magnetization.

PALEOMAGNETIC POLES AND ROTATIONAL MODEL

A virtual geomagnetic pole (VGP) was calculated for each site thought to contain a primary remanence, and all VGPs in a given age group were averaged to yield a paleomagnetic pole. Results of these calculations and statistical parameters associated with each paleomagnetic pole are given in Table 2. A comparison of the data in Table 2 with those in the compilations by *Irving and Irving* [1982], *Gordon et al.* [1984], or *May and Butler* [1986] indicates that most of the paleomagnetic poles from the Eastern Klamath terrane are displaced considerably from the apparent polar wander path for cratonic North America.

Although rigid-block rotations are common along the west coast of North America, the crudely arcuate pattern of many formations in the Klamath Mountains province and the Eastern Klamath terrane have led some workers, most recently, *Renne et al.* [1988] and *Renne and Scott* [1988], to suggest the possibility of oroclinal bending. The best evidence available from our data that can help identify the type of rotation comes from primary and secondary Permian age magnetizations described in the preceding sections. Declinations of the Permian magnetizations are shown in Figure 15, with the arrows at the northeast end of the map showing representative data from the Hawkins Creek transect. The large question mark in Figure 15 represents the portion of the Nosoni Creek transect where we encountered major overprinting of the rocks and where *Renne et al.* [1988] have reported nonrotated, Permian reversed-polarity directions. The arrow to the right of the question mark shows the declination of normal-polarity site Pi2 (inverted through the origin for clarity), and that on the left represents data from the mafic quartz diorite at the northwest end of the Nosoni Creek transect. Southwest of this transect are results from the diorite south of Shasta Lake and from the remagnetized rocks of the Mule Mountain stock.

Renne et al. [1988] conclude that contrasts between their results and those previously published are indicative of oroclinal bending of the Redding section about vertical axes. However, we believe that a primary Permian magnetization for their sites has yet to be demonstrated and that their conclusion is premature. Even if further work does provide confirmation of their data, their sites are confined to a very small area (2 km^2) and the results could most easily be explained by differential rotation of a relatively small structural block. Due to structural complexities in many areas, some differential rotation should be expected, but the remarkably consistent declinations over such a broad area (Figure 15) strongly indicate that, in general, the terrane has behaved as a single rigid block. Because of the consistency of these Permian directions and the agreement between the Jurassic results from both transects noted earlier, we will assume a rigid-block rotation of the terrane in the discussions that follow.

APPARENT POLAR WANDER

The current status of the North American apparent polar wander path for late Paleozoic/early Mesozoic time is somewhat unsatisfactory, in part because some of the older studies on which it is based were not done with tectonic applications in mind. For this reason, both the sampling and laboratory procedures may have been inadequate to characterize accurately the geomagnetic field for certain time

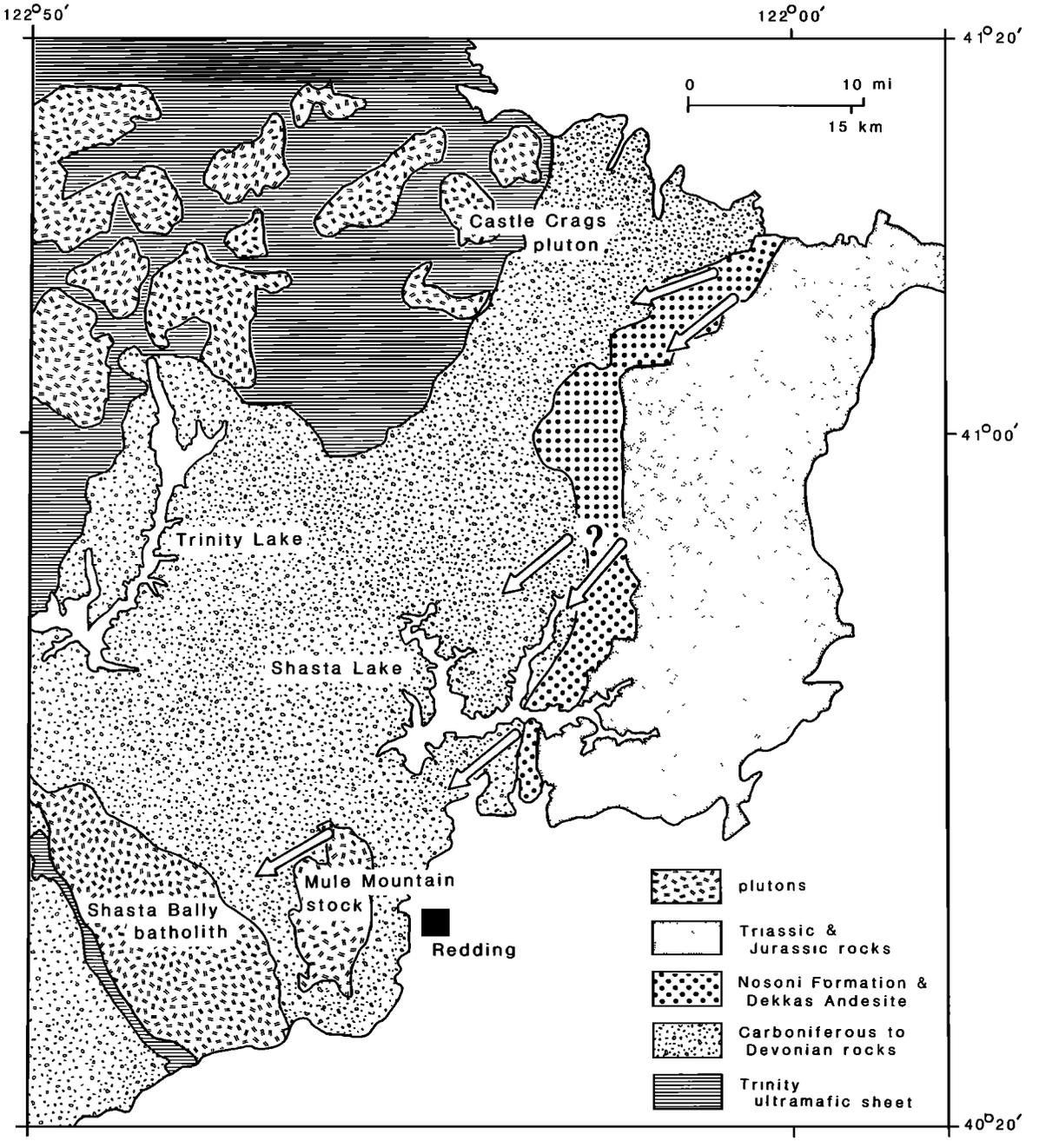


Fig. 15. Sketch of the Redding section of the Eastern Klamath terrane showing magnetic declinations (arrows) of primary or secondary magnetizations of Permian age. All declinations are from reversed-polarity sites except for site Pi2 (to the right of the question mark) whose declination is inverted through the origin for clarity. Large question mark represents portion of Nosoni Creek transect where uncertainty exists as to actual remanence directions (see text). Geology generalized mainly from *Strand* [1962].

intervals (see *Irving and Yole* [1987] for further discussion). An APW path that has been used extensively was developed by *Irving and Irving* [1982] using running means of time sequences of paleopoles with a 30-m.y. time window. However, certain portions of the APW path are based on paleomagnetic data from the Colorado Plateau which recent studies [*Bryan and Gordon*, 1986; *Steiner*, 1986, 1988] indicate may have undergone a small amount of rotation relative to the craton. *Gordon et al.* [1984] used a paleomagnetic Euler pole (PEP) analysis to construct an APW path that is similar to the *Irving and Irving* [1982] path for the period between the Permian and Middle Triassic, but one which differs considerably during the Late Triassic and Jurassic. Furthermore, as pointed out by *Irving and Yole* [1987], the method used to construct this path prejudges the nature of the processes being investigated. The most recent analysis [*May and Butler*, 1986] results in a path that differs from the other two during part of the Jurassic. This latter APW path was constructed after correcting paleopoles from the Colorado Plateau for a presumed 4° of clockwise rotation. Because of the discrepancies between published APW paths and the question of reliability of the Colorado Plateau

TABLE 3. Paleomagnetic Poles From the North American Craton

	Unit Paleopole			Mean Paleomagnetic Pole			
	PLA	PLO	A_{95}	PLA	PLO	A_{95}	Reference
Cretaceous				67.3°	186.3°	3.4°	1
Late Jurassic				...	...	...	
Canelo Hills Volcanics	62.7°	131.5°	6.3°				2
Stump Formation	64.0°	170.0°	15.0°				3
Twin Creek Limestone*	68.4°	145.0°	8.7°				4
Tuffs of Corral Canyon	61.8°	116.0°	6.2°				5
Middle Jurassic				66.9°	101.7°	2.5°	6
Barndoor intrusion	64.9°	096.7°	3.8°				7
Birdsboro intrusion	62.0°	105.0°	3.2°				8
Boyds sill	67.3°	107.4°	3.4°				9
Bridgeport dike	66.9°	095.5°	3.8°				9
Buttress dike	74.9°	092.0°	3.7°				9
Cross Rock dike	72.5°	113.9°	6.8°				9
Fairhaven dikes	68.0°	100.1°	8.5°				9
Frederick dikes	69.0°	103.7°	3.0°				9
Gettysburg intrusion	63.2°	101.5°	5.7°				8, 9
Higganum dike	65.5°	099.8°	3.3°				9
Quakertown intrusion	66.0°	098.5°	4.3°				8
York Haven intrusion	61.0°	105.5°	1.5°				8
Early Jurassic				62.7°	086.0°	5.3°	6
Deerfield Basalt†	47.0°	080.0°	8.1°				10
East Rock intrusion	70.4°	067.3°	3.6°				7
Granby Tuff†	63.2°	096.1°	6.3°				11
Hampden Basalts	73.4°	092.1°	3.8°				10
Holyoke Basalt†	58.4°	087.0°	3.8°				10, 11
Mount Carmel intrusion	66.7°	087.3°	5.0°				7
Shelburne dike	69.0°	098.0°	6.0°				12
Talcott Basalt	52.4°	078.3°	2.7°				10
Watchung Basalt†	63.5°	093.2°	6.2°				13
West Rock Diabase	63.2°	086.0°	4.9°				7
Late Triassic				58.0°	087.0°	5.7°	6
Redonda Formation†	59.5°	075.8°	9.3°				14
Chinle Formation							
Unnamed shale	56.4°	091.0°	6.0°				14
Popo Agie Formation	55.5°	095.5°	...				15
Manicouagan intrusions†	59.6°	084.1°	7.4°				16
Late Permian				52.0°	122.5°	4.5°	6
Quartermaster Formation							
Elk City Member†	53.0°	121.0°	5.0°				17
Doxey Member	47.0°	123.0°	9.0°				17
Seven Rivers Formation	54.0°	127.0°	9.0°				17
Marlow Formation	54.0°	119.0°	7.0°				17

PLA, PLO, north latitude and east longitude, respectively, of paleomagnetic pole; A_{95}, 95% confidence limit about mean pole. References: 1, *Grommé* [1984]; 2, *Kluth et al.* [1982]; 3, *Schwartz and Van der Voo* [1984]; 4, *McCabe et al.* [1982]; 5, *May et al.* [1986]; 6, this paper; 7, *Smith and Noltimier* [1979]; 8, *Beck* [1972], 9, *Smith* [1976]; 10, *de Boer* [1968]; 11, *Irving and Banks* [1961]; 12, *Larochelle and Wanless* [1966]; 13, *McIntosh et al.* [1985]; 14, *Reeve and Helsley* [1972]; 15, *Grubbs and Van der Voo* [1976]; 16, *Robertson* [1967] and *Larochelle and Currie* [1967]; 17 *Peterson and Nairn* [1971].

*Twin Creek Limestone is Middle Jurassic in age but is listed here because its remanence seems to be Late Jurassic (see text).

†Recalculated from original data.

results, we decided to reevaluate currently available data and to calculate paleomagnetic poles for the specific time intervals that are relevant to the current study. Where possible, formational or pluton means were calculated for use in the determination of a paleomagnetic pole rather than assigning unit weight to each sample locality. This procedure often resulted in poles that differed somewhat from those calculated by others, but estimates of the uncertainties about each mean obtained in this way are conservative. A conservative approach seems prudent in view of the fact that knowledge of the North American APW path is currently in a state of transition.

Permian to Triassic Data

Few paleomagnetic studies of Permian rocks from the craton exclusive of the Colorado Plateau are available and only one from the Upper Permian. *Peterson and Nairn* [1971] sampled several formations of red beds in Oklahoma and New Mexico, and the rocks were thermally demagnetized. Results from four formations were omitted in the

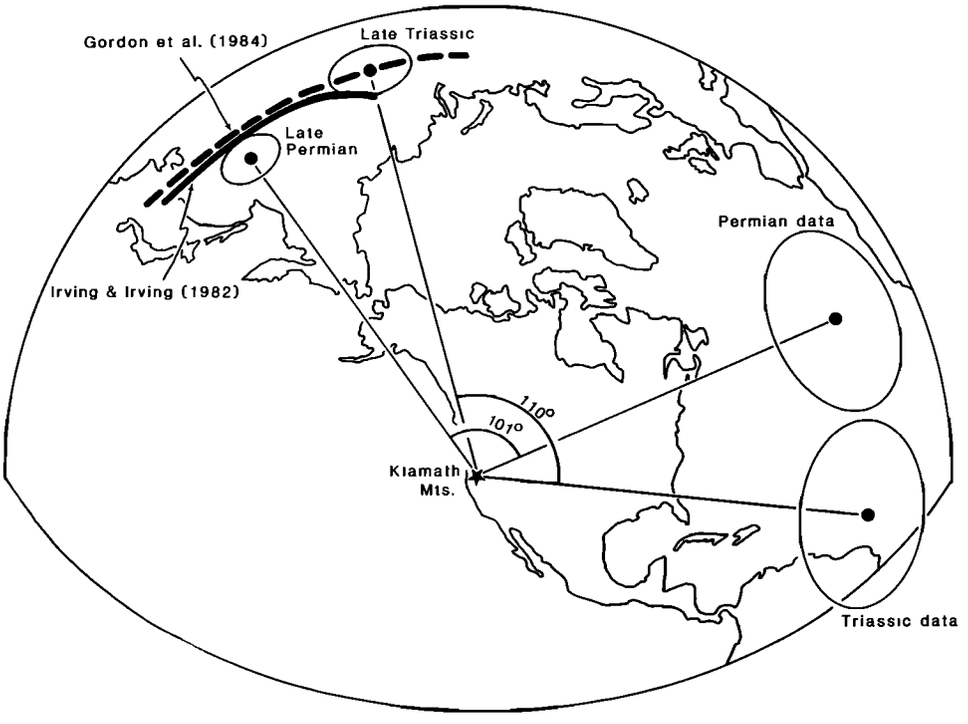


Fig. 16. Mean paleomagnetic poles and associated 95% confidence limits for Permian and Triassic rocks of the Eastern Klamath terrane. Poles are compared with reference poles calculated for Late Permian and Late Triassic time (Table 3), and portions of apparent polar wander (APW) paths determined from the Permian and Triassic by *Irving and Irving* [1982] and by *Gordon et al.* [1984].

calculation of the Late Permian pole given in Table 3 because they were based on four or fewer samples and/or they had large uncertainties (>15°) about their mean directions. Results from two of three sites within the Seven Rivers Formation and one of four within the Elk City Member of the Quartermaster Formation were also omitted from the mean formational poles for the same reason(s).

Several results are available for Early Triassic time, but few have been published for the Middle and Late Triassic. Intrusive rocks of the Manicouagan structure, Quebec, were studied paleomagnetically by *Robertson* [1967] and *Larochelle and Currie* [1967]. Results from units 1, 2, and 4 of *Larochelle and Currie* [1967] have been combined with those from sites 3, 4, 6, 7, and 8 of *Robertson* [1967] to yield the Manicouagan pole listed in Table 3. There may be some duplication between the two studies, and *Robertson* [1967] did not indicate whether any of his sites may have been in the same intrusion. However, the mean Manicouagan pole in Table 3 should be less biased than if the results from all 19 sampled sites had been used in the calculation. The pole for the Redonda Formation was calculated using the results from the upper and middle polarity zones described by *Reeve and Helsley* [1972], and their result from the unnamed shale member of the Chinle Formation is listed separately. The entry for the Popo Agie Formation in Table 3 is based on the results from one site on the craton adjacent to the Idaho-Wyoming overthrust belt [*Grubbs and Van der Voo*, 1976; *Van der Voo and Grubbs*, 1977].

The paleomagnetic poles just calculated are far from ideal, but refinements must wait until new data become available for these specific time periods. These poles, however, do not differ significantly from the APW paths developed by either *Irving and Irving* [1982] or *Gordon et al.* [1984] (Figure 16), indicating that they are not grossly in error and that they are adequate for comparison with the Eastern Klamath data.

Jurassic Data

The Jurassic APW path for North America is poorly constrained both because this was a period of rapid APW and because of a paucity of reliable data. The three recently constructed APW paths discussed above differ considerably for these reasons, and we further diminish the data base by limiting our discussion to localities exclusive of those on the Colorado Plateau. Readers should refer to the analysis by *May and Butler* [1986] for a thorough review of all available paleomagnetic results from Jurassic rocks of North America.

Late Triassic to Early Jurassic redbeds of the Newark Supergroup in New Jersey were initially studied paleomagnetically by *Opdyke* [1961], who showed that no significant error in magnetic inclination occurs in these sediments. Recently, *McIntosh et al.* [1985] demonstrated that although the sediments are suitable for magnetostratigraphic studies, most retain some secondary remanence components after magnetic cleaning and should not be used for tectonic reconstructions. Slightly younger than the sediments are the igneous rocks of the Newark Supergroup that are exposed throughout much of northeastern North America. A comprehensive review of all paleomagnetic studies of these igneous rocks was made by *Smith and Noltimier* [1979], who found that the mean remanence directions of the rocks whose ages clustered at about 190 Ma differed significantly from those of the younger group that have ages about 175 Ma. We here recalculate mean paleomagnetic poles for the 190 and 175 Ma groups, basically as defined by *Smith and Noltimier* [1979],

but using a single direction calculated for an individual intrusion or a basalt formation and assigning unit weight to each of these means. By doing so, any bias toward directions recorded by those units that were the most thoroughly sampled should be eliminated, and the resulting paleomagnetic poles should have realistic uncertainty limits.

Included in the Early Jurassic (≃ 190 Ma) pole in Table 3 are the results from the Watchung (name now abandoned), Talcott, Hampden, Holyoke, and Deerfield basalts, the Granby Tuff, the West Rock Diabase and the East Rock, and Mount Carmel intrusions. Because the initial results from the Watchung basalts (*Opdyke*, 1961] were based on only two or three independently oriented samples from each site, the Watchung mean in Table 3 is based on the results of *McIntosh et al.* [1985] for sites where five or more samples were obtained. Although *Smith and Noltimier* [1979] assigned the Shelburne dike of Nova Scotia to the 175 Ma group based on its paleomagnetic direction, its 197 ± 16 Ma (1 s.d.) age [*Larochelle and Wanless*, 1966] indicates that it belongs to the older group, and we include its direction in the calculation of the 190 Ma reference pole. The North Mountain basalts of Nova Scotia have yielded K-Ar ages of 200 ± 10 Ma [*Carmichael and Palmer*, 1968] but *Smith and Noltimier* [1979] did not include the mean magnetization direction for these flows in their calculations because the results differed from others of the same age group. Because at least 16 flows constitute the North Mountain basalts, geomagnetic secular variation should be more nearly averaged for this unit than for most of the basalt formations of the Newark trend. However, the results reported by *Larochelle* [1967] and by *Carmichael and Palmer* [1968] were based on one or two samples from each site, and the directions therefore may not always have been accurate representations of the geomagnetic field at the time a particular lava flow erupted. For this reason, we also omit the results from the North Mountain basalts in the calculation of the 190 Ma reference pole.

The Middle Jurassic (≃ 175 Ma) pole in Table 3 includes data from the postfolding intrusions of Connecticut, the Boyds sill and the Frederick dikes of Maryland, and the intrusions in the Gettysburg Basin of southeastern Pennsylvania. The diabase intrusions of New Jersey and Pennsylvania [*Opdyke*, 1961] were assigned to the younger group of igneous rocks by *Smith and Noltimier* [1979] on the basis of their paleomagnetic directions, but these results are omitted here because of the lack of age determinations.

The Early and Middle Jurassic reference poles that we calculate (Table 3) are, not surprisingly, indistinguishable from those calculated by *Smith and Noltimier* [1979]. However, the use of a smaller N for both the older (10 formations or intrusions rather than 72 sites) and younger (11 cooling units versus 156 sites) groups of rocks yields what should be more meaningful uncertainty limits. *Irving and Yole* [1987] used results from some of the Newark Supergroup igneous rocks and combined them with coeval data from Africa rotated into the common framework of Pangaea to calculate a Lower Jurassic reference paleopole. Their paleopole (68°N, 095°E) differs from ours (62.7°N, 086.0°E) by 6.5° of arc, and the difference is statistically significant. Although discrepancies still exist because of how the various data are interpreted, it should be remembered that probably none of the poles are perfectly representative of the Early and Middle Jurassic geomagnetic fields. Many of the basalts or basalt formations used in the calculations consist of single flows or a very small number of independent cooling units and the effects of geomagnetic secular variation were probably not averaged. Secular variation probably has been averaged in some of the larger intrusions, but how many independent recordings of the geomagnetic field have been sampled is unknown. For example, there is some evidence to suggest that the East Rock, West Rock, and Mount Carmel bodies may be exposures of a single large intrusive sheet [*Smith and Noltimier*, 1979].

The remaining four Jurassic poles in Table 3 are those calculated for individual studies. The ash flow tuffs thought to be part of the Canelo Hills Volcanics [*Kluth et al.*, 1982] were considered to belong to part of the Glance Conglomerate by *May and Butler* [1986]. However, because the Glance Conglomerate is still considered to be Early Cretaceous in age, we will continue to refer to the "Canelo Hills Volcanics" paleopole. Biostratigraphic ages on the Stump Formation [*Schwartz and Van der Voo*, 1984] and Twin Creek Limestone [*McCabe et al.*, 1982] indicate that they should be intermediate between, and partially overlap, the age ranges of the Canelo Hills Volcanics and the welded ash flow tuffs of Corral Canyon [*May et al.*, 1986]. However, because sedimentary sequences and the time of their remanence acquisition often cannot be precisely dated, and the mean directions themselves are readings of the geomagnetic field averaged over varying and often considerable lengths of time, the differences in the poles shown in Figure 17 may be real. Note that some localities in both the Twin Creek Limestone and the Stump Formation are in the Idaho-Wyoming overthrust belt but apparently have not been significantly rotated [*McCabe et al.*, 1982; *Schwartz and Van der Voo*, 1984] in contrast to Early Triassic rocks in the belt [*Grubbs and Van der Voo*, 1976]. Even though there are questions as to the reliability of these two poles, they are shown in Figure 17 as possible constraints to the approximate APW path, which must necessarily remain questionable for the Late Jurassic. The dashed part of the APW curve in Figure 17 during the latest Triassic and earliest Jurassic period shows the "cusp" predicted by the PEP analyses by *Gordon et al.* [1984] and *May and Butler* [1986].

Comparison With Reference Paleopoles

Comparison of results from the Permian of the Eastern Klamath terrane with the Late Permian reference pole (Figure 16) indicates a clockwise rotation of 101.1° ± 10.1° for the Permian rocks. Uncertainties given in this report are 95% confidence limits calculated by the method of *Demarest* [1983]. A comparison with the Triassic reference pole (Figure 16) indicates a clockwise rotation of 110.3° ± 12.0° for the Triassic strata. The Permian data indicate a small and statistically insignificant amount of southward displacement for the terrane (0.2° ± 9.6°), as do the Triassic data (6.7° ± 11.4°). Despite the fact that the strata sampled range in age from Middle Permian to Late Triassic and that the reference paleomagnetic poles are not well defined, the similarity in the amounts of indicated rotation shows that both sequences rotated together and that substantial rotation could not have begun before the latter part of Late Triassic time. Combining both data sets yields a post-Triassic, clockwise rotation of 105.7° ± 12.2°.

Although the paleomagnetic poles for the Permian and Triassic strata of the Eastern Klamath terrane (Table 2) are appropriate for estimating tectonic rotation, some of the

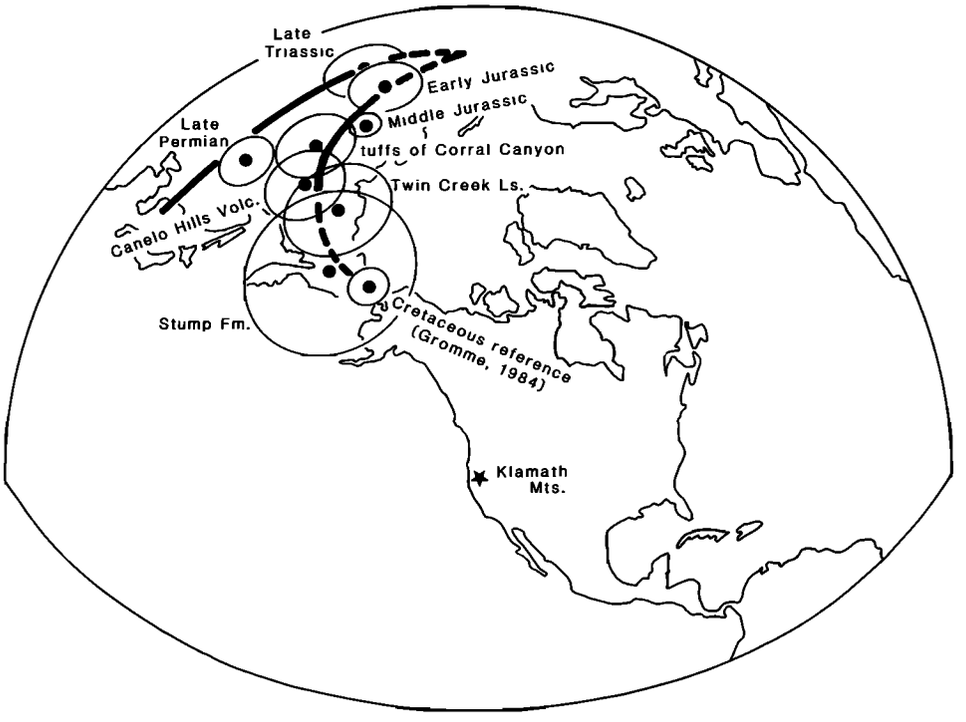


Fig. 17. Mean paleomagnetic poles and associated 95% confidence limits for studies used to constrain the APW path for Jurassic time (Table 3). Dashed portions are for time intervals where the path is poorly defined (see text).

measured dispersion could be due to small amounts of relative rotation about vertical axes between the sampling sites. Such relative rotations have been shown to be quite common in tectonically disrupted terranes. In order to obtain more precise estimates of possible latitudinal displacements, we transformed site mean inclinations to virtual geomagnetic latitudes and used the method of *McFadden and Reid* [1982] to determine mean paleolatitudes and 95% confidence limits for these strata. Inclinations measured in the Permian and Triassic rocks yield mean paleolatitudes of 18.1° ± 7.0° and 18.3° ± 9.7°, respectively. Comparing these values with expected Permian (18.5° ± 4.5°) and Triassic (11.9° ± 5.7°) paleolatitudes for stable North America, we again note statistically insignificant amounts of latitudinal displacement. Combining the Permian and Triassic results indicates a statistically insignificant amount of southward transport (3° ± 9°) relative to the craton.

Comparison of the Jurassic results of the Eastern Klamath terrane with the Jurassic APW path indicates significant clockwise rotations for two groups of directions and none for the third (Figure 18*a*). The paleopole coincident with the APW path is in marked contrast to all other data obtained during this study, which implies that it is the youngest and was acquired after all rotation had occurred. As previously noted, however, this paleopole was calculated from the magnetization direction recorded by sites that are in intermediate positions, both stratigraphically and topographically. Because this direction was found in both transects and because the data pass the fold test, a prefolding and, presumably, primary magnetization is indicated. A remagnetization of only part of these strata would not be expected because similar lithologies were sampled at all Arvison Formation sites. Moreover, remagnetization would have to be extremely selective not to affect underlying and overlying strata in either transect. The simplest explanation is that the geomagnetic field was accurately recorded and that a complex rotational history, including some retrograde motion, is indicated for the terrane. However, it also is difficult to have complete confidence in this explanation because the intermediate sites must have undergone the same clockwise rotation that affected the youngest group of sites, making the location of their paleopole on the APW path purely coincidental.

Paleomagnetic poles for units whose directional data are shown in Figure 12 also are compared with the Jurassic APW path (Figure 18*b*). Despite several possible explanations [*Schultz*, 1983] for the concordance of the in situ remanence direction of the Ashland pluton with the expected direction, *Schultz* [1983] and *Schultz and Levi* [1983] consider the Ashland pluton results as problematic. Although a later remagnetization of this pluton might be suspected, there is no firm evidence to suggest that its remanence is any different from that recorded by the other plutons studied, and the fact that both polarities were found in the Ashland adds support for a primary magnetization. We suggest that its remanence may have been acquired at nearly the same time as that recorded by the three sites within the Arvison (Ar2, Ar3, Ar4) which also yield directions close to the expected direction. If this interpretation is correct, the acquisition time of this direction can be dated by the isotopic age of the pluton which is about 160 Ma [*Lanphere et al.*, 1968; *Hotz*, 1971]. While the direction recorded by the Ashland pluton may or may not be equivalent to that in the intermediate Arvison sites, the Jurassic overprint reported by *Renne et al.* [1988] has a direction (Figure 18*b*) that is nearly identical and most likely was acquired during the same time interval. These authors consider the overprint to be Middle Jurassic (pre-168 Ma) in age, which agrees with our estimate. However, the overprint that they report is postfolding, while the direction that we report is prefolding.

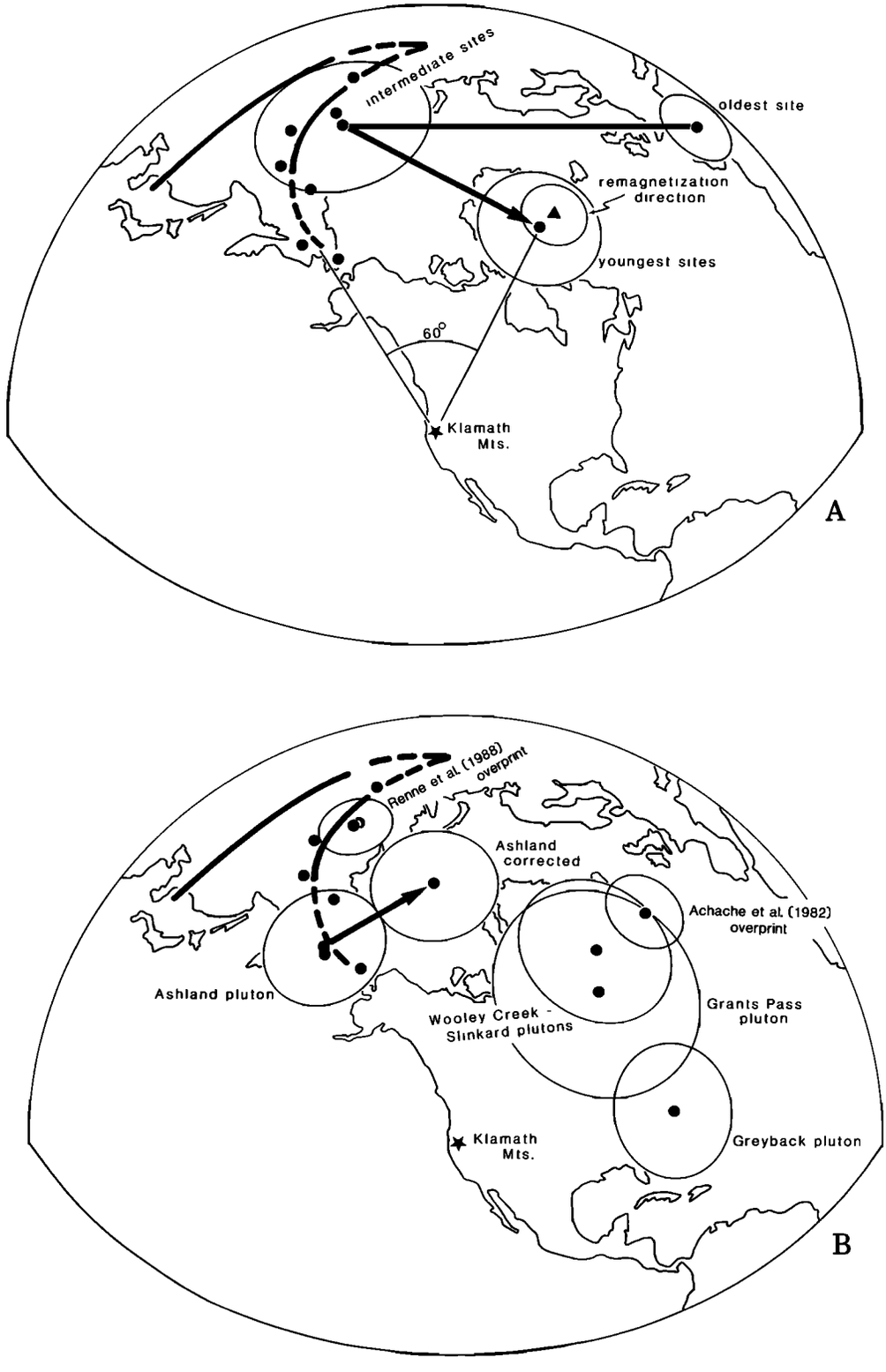


Fig. 18. Comparison between APW path shown in Figure 17 and mean poles and associated 95% confidence limits for data from (*a*) Jurassic strata of Eastern Klamath terrane sampled during this study and (*b*) from Jurassic plutons in the composite Western Paleozoic and Triassic terrane [*Schultz*, 1983], and overprint directions [*Achache et al.*, 1982; *Renne et al.*, 1988] within the Eastern Klamath terrane that also may be Jurassic in age. Confidence ovals of reference Jurassic data are omitted for clarity.

Thus the significance of this magnetization direction remains unclear.

Agreement between the magnetization directions of the Grants Pass and Wooley Creek-Slinkard plutons and our youngest Jurassic strata indicates that the magnetization of the latter was acquired in the interval between about 140 and 155 Ma [*Lanphere et al.*, 1968; *Hotz*, 1971; *Irwin*, 1985]. Although recent evidence [*Irwin*, 1985; *Barnes et al.*, 1986*a*;

Wright and Fahan, 1988] indicates that the Wooley Creek and Slinkard plutons have emplacement ages close to that of the Ashland pluton, *Barnes et al.* [1986*a*] suggest that the approximately 155 Ma K-Ar ages for the Wooley Creek and Slinkard [*Lanphere et al.*, 1968] may reflect either slow cooling of the plutons or a later thermal event that has reset their K-Ar systems. The magnetization of these plutons could therefore be younger than that of the Ashland pluton. Ages of approximately 160 Ma and 150 Ma for the intermediate and youngest groups, respectively, of Jurassic directions recorded by our Eastern Klamath sample localities are allowed by the large uncertainties in the paleontologically determined ages of the Arvison and Potem Formations. When all evidence is considered, we believe that our initial interpretation of a primary magnetization for these strata is valid. Because the oldest Jurassic direction from the Eastern Klamath terrane is from a single site whose stratigraphic position is uncertain, a numerical value for its apparent clockwise rotation is not calculated. Evaluation of the youngest direction is hampered because of the lack of a latest Jurassic reference pole. However, the mean direction of the youngest site would coincide with the approximately determined APW path if 60° of clockwise rotation of the terrane were removed (Figure 18*a*). A pole calculated from this mean direction would lie approximately 15° of arc west of the Cretaceous reference pole of *Grommé* [1984]. Such a location would be approximately correct for a paleomagnetic pole slightly older than the oldest datum (135 Ma) used to calculate the Cretaceous reference pole if no latitudinal displacement of the terrane had occurred.

Discussion and Conclusions

In summary, these paleomagnetic results show that many of the Permian through Jurassic strata of the Eastern Klamath terrane possess a magnetization that is both prefolding and most likely primary. Similar amounts of clockwise rotation ($\simeq$ 106°) relative to stable North America have been found for both the Permian and Triassic rocks. The initial rotation of the terrane may have been completed prior to Middle Jurassic time if the data from our intermediate Jurassic sites (circa 160 Ma) are reliable. Data from our youngest Jurassic sites again indicate clockwise rotations of about 60° for the terrane during Late Jurassic time (circa 150 Ma). The only available paleomagnetic information about Klamath Mountain rocks younger than the Jurassic are from superjacent Cretaceous and Tertiary strata [*Mankinen and Irwin*, 1982] and from plutons of the Shasta Bally belt, the youngest (Early Cretaceous) belt of postamalgamation plutons in the province [*Mankinen et al.*, 1988]. The superjacent strata, although largely remagnetized, indicate no more than 14.4° ± 6.9° of clockwise rotation for the province since the Cretaceous [*Mankinen et al.*, 1988]. Results from the Shasta Bally plutonic belt (25.7° ± 13.6° of clockwise rotation) indicate that rotation of the province was nearly complete by about 136 Ma. No latitudinal displacements are required by either these data or those from the older rocks.

At least three episodes of remagnetization prior to the late Tertiary have been recognized in the Klamath Mountains province. The earliest episode recognized was a prefolding remagnetization of the Mule Mountain stock during the Permian. A second, postfolding overprint affected some of the Permian rocks at the Hawkins Creek and Nosoni Creek transects and the Pit River stock south of Shasta Lake. The coincidence of this second overprint direction with remanence directions recorded by some of our Jurassic sites and some of the plutons sampled by *Schultz* [1983] indicates that it probably occurred during the Late Jurassic. If this interpretation is correct, the older rocks of both transects were folded prior to this time and were not subsequently involved in later deformation affecting the Arvison and Potem formations. The youngest episode was a postfolding overprint found in rocks near the Nosoni Creek transect west of the McCloud River arm of Shasta Lake. This overprint seems to have occurred during the same period when much of the onlap Great Valley sequence south of the province was remagnetized. *Mankinen and Irwin* [1982] considered this remagnetization to be no younger than latest Cretaceous or earliest Tertiary.

Other Investigations

Paleomagnetic samples were obtained from 21 sites within limestone units of the Devonian Kennett Formation in the Eastern Klamath terrane by *Achache et al.* [1982]. The remanence of 10 of these units was found to be a stable postfolding remagnetization that these authors relate to the intrusion of the terrane by the Shasta Bally batholith at about 136 Ma. This interpretation requires 70° ± 11° of clockwise rotation and a few hundred kilometers of northward translation for the terrane since the Early Cretaceous [*Achache et al.*, 1982]. The work of *Mankinen et al.* [1988], however, shows that the direction of the Shasta Bally batholith is markedly dissimilar to that recorded by the Kennett and that this pluton could not have been the cause of the remagnetization. Although the Permian Pit River stock is located within 8 km of the sample localities, the directions of the Kennett and those of the Permian rocks sampled for this study also differ. We believe that the remagnetization most likely occurred during the Late Jurassic (note close correspondence with paleopoles in Figure 18*b*) which would make the data obtained by *Achache et al.* [1982] entirely compatible with our own and those of *Schultz* [1983].

Paleomagnetic samples were obtained from seven sites within the Redding section of the Eastern Klamath terrane by *Fagin and Gose* [1983]. Results from four of the sites, which range in age from Devonian to Late Triassic, were considered to be reliable. Results from two of the sites were discussed above. The remaining two sites thought to preserve a primary magnetization were in the Kennett (Devonian) and Bragdon (Mississippian) formations, the results from which have uncertainty limits of 28.1° and 15.1°, respectively. The similarity in the remanence directions of these two sites with that of the much younger Modin Formation (Triassic age) would not be expected if the magnetization of these strata was indeed primary, although too little information is provided to allow a proper assessment of their results.

Paleomagnetic samples were obtained from metamorphosed volcaniclastic rocks of the Upper Jurassic Galice Formation near the Rogue River, Oregon, in the Western Jurassic terrane by *Bogen* [1986*a*]. Samples from 13 sites have a stable secondary magnetization that postdates folding of the rocks. A clockwise rotation of about 100° is indicated for the sites since the rocks were remagnetized. Unless there are undetected structural complications, larger amounts of clockwise rotation seem to be indicated for the northern part of the Western Jurassic terrane than for the Western Paleo-

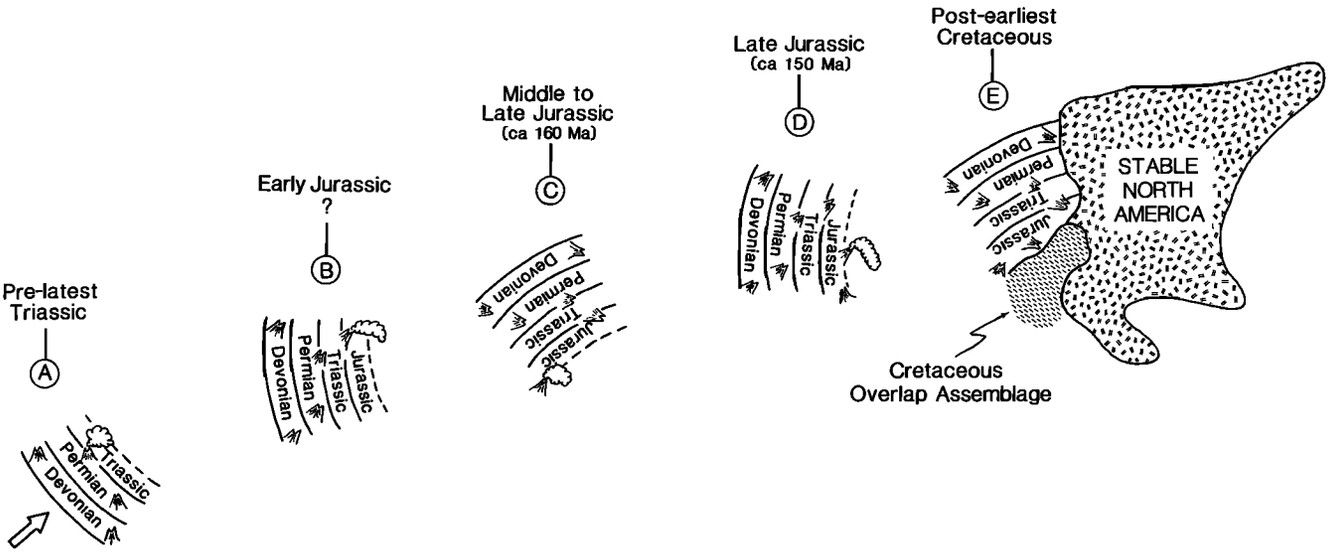


Fig. 19. Schematic diagram showing sequence of rotation of volcanic island arc represented by Redding section of the Eastern Klamath terrane and its accretion to North America. No latitudinal displacement can be detected by the available paleomagnetic data. Modified from *Irwin* [1989].

zoic and Triassic, and the Eastern Klamath terranes during the same interval. The possibility of such structural complications has been recently discussed by *Renne and Scott* [1986], *Harper and Park* [1986], and *Bogen* [1986*b*].

Accretionary Model

Much additional data must be obtained before the complex tectonic history of the Klamath Mountains province can be fully understood. However, a general sequence of events can be deduced for the Eastern Klamath terrane which formed the primitive nucleus of the Klamath Mountains province. Although the original orientation of the pre-Permian rocks of the Redding section is unknown, the current arcuate pattern of the formations, and perhaps the edge of the original island arc, was facing toward the southwest during Late Paleozoic time (Figure 19*a*). The various elements of the western Klamath Mountains were swept against this terrane from the southwest or west since Devonian time when the Central Metamorphic terrane was amalgamated [*Irwin*, 1981], providing some evidence that the present outcrop distribution and the original arc faced roughly in the same direction. The similarity of paleomagnetic directions from Permian and Triassic strata of the arc shows that major rotation of those strata could not have begun until latest Triassic or earliest Jurassic time. Perhaps the rotation began as the arc neared the continental margin and, once started, continued as the Lower and Middle Jurassic strata were being deposited. This initial rotation may have been completed during the Early or Middle Jurassic (Figures 19*b* and 19*c*), after which time some retrograde motion occurred and the arc was again facing west by about 150 Ma (Figure 19*d*). Similarities in magnetization directions between the various units (Figure 18) also indicate that the Western Paleozoic and Triassic rocks, the Eastern Klamath terrane, and the intervening Central Metamorphic terrane were already amalgamated by Middle to Late Jurassic time and underwent retrograde motion and subsequent rotation as a single block. Indeed, the suturing of the Western Paleozoic and Triassic rocks to the Central Metamorphic terrane has been considered a Middle Jurassic event on the basis of various criteria [*Irwin*, 1981]. Geologic evidence also indicating amalgamation of the Western Jurassic terrane by latest Jurassic or earliest Cretaceous time [*Irwin*, 1981] further demonstrates the need for additional data to assess fully the significance of what seems to be an excessive amount of rotation as noted by *Bogen* [1986*a*]. In any case, the Klamath Mountains terranes continued rotating until they attained essentially their present northwest facing by the Early Cretaceous (Figure 19*e*). The close correspondence between the last stages of rotation and the inception of deposition of the Lower Cretaceous basal strata of the Great Valley sequence on Klamath basement probably represents completion of accretion to the North American continent. Most of the rotation of the terranes of the Klamath Mountains occurred while the terranes were parts of oceanic plates, but minor rotation may have occurred later during postaccretionary extension in the Basin and Range province [*Mankinen et al.*, 1988].

Relationship of the Arc to Other Elements of the Cordillera

At present, there is no general consensus as to the specific paleogeographic relation of the Klamath Mountains province to other rocks of the western Cordillera. The original location of the island arc that formed the Eastern Klamath terrane is unknown because longitudinal displacements cannot be detected by the paleomagnetic method and because paleontologic evidence for displacement is equivocal. Transport of the arc as much as several thousand kilometers from the west could have been accomplished without significant latitudinal displacement because convergence between the Farallon plate and the western margin of North America, although poorly constrained, appears to have been predominantly east-west during Middle Jurassic to Early Cretaceous time [*Engebretson et al.*, 1985]. A few Tethyan fossils have been reported from other areas within the Klamath Moun-

tains, but typical Tethyan faunal assemblages have not been found in the Eastern Klamath terrane so large-scale transport of the arc did not necessarily occur. The limestone masses within the Dekkas Andesite that contain a waagenophyllid coral species also contain many fossils with non-Tethyan affinities [*Stevens et al.*, 1987]. This association has been interpreted as indicating that some Tethyan species were able to migrate via a network of Permian seamounts well into the paleo-Pacific Ocean [*Miller and Wright*, 1987]. However, based on the paleomagnetic data, the Permian and Triassic arc must have been at a high angle to the trend of the continental margin, which indicates that the arc probably was not close to the North American continent. It is also apparent that the arc had to originate sufficiently far offshore in order to have developed distinctive Permian fusulinid and coral faunas that differ from typical North American forms [*Stevens*, 1977]. For other marine fossils as well, the occurrence of large numbers of forms not reported elsewhere (see, for example, *Sanborn* [1960]) suggests that the volcanic arc was an isolated environment. Despite the distinctive nature of the Eastern Klamath fossil fauna, some affinity to forms along the western edge of the craton has also been noted (see, for example, *Boucot and Potter* [1977] and *Wilson* [1982]) perhaps indicating that the ocean basin separating the two areas may not have been very wide. Favoring a separation of hundreds, rather than thousands, of kilometers between the province and North America, *Miller* [1989] has cited and summarized arguments indicating that the Eastern Klamath volcanic arc (or a similar terrane) could have been part of the source area for Permian volcaniclastic strata in the Golconda allochthon to the east in west central Nevada.

Geochemical characteristics of the Early Devonian volcanic sequence that forms the roots of the Eastern Klamath arc, however, were considered by *Lapierre et al.* [1985] and *Brouxel et al.* [1987] to be representative of an immature island arc that was in a remote position relative to the Devonian continents. Oxygen isotopic data [*Taylor and South*, 1985] also show no evidence for the presence of continental crust during generation of silicic magmas of the terrane, although lead isotope compositions [*Doe et al.*, 1985] are more equivocal.

The location of the continental margin at the time the Klamath Mountains terranes were accreted also is not clearly known. Some researchers have suggested that the province may have been east of its present longitude during and after accretion and subsequently moved westward during Basin and Range extension (see summary by *Grommé et al.* [1986]). One attractive feature of a more easterly location for the province is that it would have been out of the way of the terranes that were being transported northward [*Beck et al.*, 1981; *Irving et al.*, 1985] during the Late Cretaceous and early Tertiary. The strata of the Eastern Klamath terrane would also have been much closer to the allochthonous Permian and Triassic rocks near Quinn River Crossing in northwestern Nevada. Fusulinids that are characteristic of the McCloud Limestone were found in a limestone sequence at Quinn River Crossing by *Skinner and Wilde* [1966], who thought that the two areas must have been connected during Early Permian time. Striking similarities in both faunal assemblages and lithology between the two areas are described further by *Ketner and Wardlaw* [1981] and *Silberling and Jones* [1982]. Unfortunately, preliminary paleomagnetic measurements from a locality of Permian limestone at Quinn River Crossing suggest that these rocks may have been involved in the large-scale, Late Cretaceous to early Tertiary remagnetization found in other rocks in the area [*Russell et al.*, 1982].

A third consequence of a more easterly position would be a closer proximity between the Klamath Mountains province and the Sierra Nevada. Because similar rock sequences of roughly the same age occur in both areas, various researchers have tried to establish definitive correlations between the two with limited success. Paleontologic data, as summarized above, indicate that a distinctive fossil fauna was developed in the Eastern Klamath terrane, whereas many assemblages from the Sierra Nevada appear to have typical North American affinities (see summary by *Hannah and Moores* [1986]). Although *Hannah and Moores* [1986] report that new paleontologic and stratigraphic data seem to indicate some links between contemporaneous rocks in the Klamath Mountains, the Sierra Nevada, and the Great Basin, they note also that the possible correlations remain tentative.

Paleomagnetic data from Cretaceous plutons in the Sierra Nevada show concordance with expected pole positions [*Grommé et al.*, 1967; *Frei*, 1986] in contrast to those from the rotated plutons of the Klamath Mountains, which indicates that the two provinces do not share a common post-Jurassic tectonic history. Geochemical data have been presented by *Lapierre et al.* [1987], who point out significant differences between nearly contemporaneous magmatism during Paleozoic and early Mesozoic time in the Klamath Mountains and the Sierra Nevada. Finally, seismic refraction data [*Zucca et al.*, 1986; *Fuis et al.*, 1987] seem to indicate the presence of granitic and metamorphic rocks similar to those in the Sierra Nevada beneath the volcanic and sedimentary rocks of the Modoc Plateau. If the interpretation of the seismic data is correct, accretion of the Klamath Mountains terranes significantly to the east of their present location seems unlikely.

Two other possible correlatives to the Eastern Klamath terrane are briefly noted, the first being the section of pre-Tertiary rocks in the Blue Mountains area of eastern Oregon. Paleomagnetic data from Late Jurassic and Early Cretaceous plutons in this province indicate 60° ± 29° of clockwise rotation and no detectable latitudinal displacement [*Wilson and Cox*, 1980]. Here most of the rotation is pre-Eocene in age [*Grommé et al.*, 1986] and, like the rotations in the Klamath Mountains, clearly preceded the Tertiary rotation of the Oregon Coast Range. *Grommé et al.* [1986] consider the pre-Eocene part of the rotations observed in the Blue Mountains plutons and in the Seven Devils Group and Huntington Formation [*Hillhouse et al.*, 1982] to have occurred during the last stages of accretion of these terranes to the continental margin. Another possible correlation is with rocks of the Stikine terrane in British Columbia where Permian fusulinids similar to those in the McCloud Limestone have been found [*Ross and Monger*, 1978; *Ross and Ross*, 1983]. Paleolatitudes determined for volcanic rocks of the Lower Permian Asitka Group in the Stikine terrane are similar to those from the Dekkas Andesite in the Eastern Klamath terrane, and *Irving and Monger* [1987] suggested that the two areas were much closer together in Permian time. *Wernicke and Klepacki* [1988] have presented a model that explains the displacement of rocks comprising the Stikine terrane from their original position (in what became the Columbia embayment) as a result of the

collision and penetration of Wrangellia into North America beginning in Middle Jurassic time. The fact that no paleolatitude anomaly is found in rocks of the Eastern Klamath terrane places the entire burden of departure on Stikinia.

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